Base R: Exercises with Solutions

Outline

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Vectors and Types (Ex 1–18)

Ex 1: Logical Indexing

Problem

Let `v <- c(14, -3, 22, 7, -8, 0, 31, 5, -1, 18)`.

- Extract all elements strictly between 5 and 25
- Replace every negative element with 0 in a copy `w`
- How many elements are negative or exceed 20?
- Find indices where $|v_i|$ exceeds its rank

Sol 1

```r
v <- c(14, -3, 22, 7, -8, 0, 31, 5, -1, 18)
v[v > 5 & v < 25]
```

```
# [1] 14 22  7 18
```

```r
w <- v; w[w < 0] <- 0; w
```

```
#  [1] 14  0 22  7  0  0 31  5  0 18
```

```r
sum(v < 0 | v > 20)
```

```
# [1] 5
```

```r
which(abs(v) > rank(v))
```

```
# [1]  1  2  3  4  5  7 10
```

Ex 2: Named Vectors

Problem

Create `stock <- c(AAPL=189.5, MSFT=415.2, GOOG=172.3, AMZN=185.7)`.

- Access `GOOG` by name; add `NVDA = 875.4`
- Compute percentage change from base 150; round to 1 dp
- Return stocks above mean price, sorted descending
- Rename all tickers to lowercase

Sol 2

```r
stock <- c(AAPL=189.5, MSFT=415.2, GOOG=172.3, AMZN=185.7)
stock["GOOG"]; stock["NVDA"] <- 875.4
```

```
#  GOOG
# 172.3
```

```r
round((stock - 150) / 150 * 100, 1)
```

```
#  AAPL  MSFT  GOOG  AMZN  NVDA
#  26.3 176.8  14.9  23.8 483.6
```

```r
sort(stock[stock > mean(stock)], decreasing=TRUE)
```

```
#  NVDA  MSFT
# 875.4 415.2
```

```r
names(stock) <- tolower(names(stock)); names(stock)
```

```
# [1] "aapl" "msft" "goog" "amzn" "nvda"
```

Ex 3: `which`, `which.min`, `which.max`

Problem

Let `x <- c(3, 7, 2, 9, 1, 5, 8, 4, 6, 10)`.

- Indices of elements > 6
- Index of min and max
- Second-largest value (no `sort`)
- Local maxima: elements greater than both immediate neighbours (ignore endpoints)

Sol 3

```r
x <- c(3, 7, 2, 9, 1, 5, 8, 4, 6, 10)
which(x > 6)
```

```
# [1]  2  4  7 10
```

```r
which.min(x); which.max(x)
```

```
# [1] 5
# [1] 10
```

```r
max(x[x != max(x)])
```

```
# [1] 9
```

```r
n <- length(x); idx <- 2:(n-1)
x[idx][x[idx] > x[idx-1] & x[idx] > x[idx+1]]
```

```
# [1] 7 9 8
```

Ex 4: Cumulative and Parallel Functions

Problem

Let `x <- c(2,5,3,8,1,7,4)` and `y <- c(6,2,9,1,4,3,8)`.

- Running maximum of `x`
- Running sum and product
- Element-wise min and max
- Verify $\sum \text{pmin}(x, y) + \sum \text{pmax}(x, y) = \sum x + \sum y$

Sol 4

```r
x <- c(2,5,3,8,1,7,4); y <- c(6,2,9,1,4,3,8)
cummax(x)
```

```
# [1] 2 5 5 8 8 8 8
```

```r
cumsum(x); cumprod(x)
```

```
# [1]  2  7 10 18 19 26 30
# [1]    2   10   30  240  240 1680 6720
```

```r
pmin(x,y); pmax(x,y)
```

```
# [1] 2 2 3 1 1 3 4
# [1] 6 5 9 8 4 7 8
```

```r
sum(pmin(x,y)) + sum(pmax(x,y)) == sum(x) + sum(y)
```

```
# [1] TRUE
```

Ex 5: Set Operations

Problem

Let `a <- c(1,3,5,7,9,3,5)` and `b <- c(2,3,5,6,7,2)`.

- Unique elements of `a`
- Positions of duplicated values in `a`
- $a \cup b$, $a \cap b$, $a \setminus b$
- `match(c(3,5,4), b)` — explain the `NA`

Sol 5

```r
a <- c(1,3,5,7,9,3,5); b <- c(2,3,5,6,7,2)
unique(a)
```

```
# [1] 1 3 5 7 9
```

```r
which(duplicated(a))
```

```
# [1] 6 7
```

```r
union(a,b); intersect(a,b); setdiff(a,b)
```

```
# [1] 1 3 5 7 9 2 6
# [1] 3 5 7
# [1] 1 9
```

```r
match(c(3,5,4), b)  # 4 not found -> NA
```

```
# [1]  2  3 NA
```

Ex 6: Recycling Rule

Problem

- What does `c(1,2,3,4,5,6) + c(10,20)` return? Why?
- Let `m <- matrix(1:12, nrow=3)`. What does `m + c(100,200,300)` do?
- Extract odd-indexed and even-indexed elements of `1:12` using `c(TRUE,FALSE)`
- Generate `1 0 1 0 1 0 1 0 1 0` using `rep`

Sol 6

```r
c(1,2,3,4,5,6) + c(10,20)                          # recycled
```

```
# [1] 11 22 13 24 15 26
```

```r
m <- matrix(1:12, nrow=3)
m + c(100,200,300)    # adds per column (col-major)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]  101  104  107  110
# [2,]  202  205  208  211
# [3,]  303  306  309  312
```

```r
x <- 1:12; x[c(T,F)]; x[c(F,T)]
```

```
# [1]  1  3  5  7  9 11
# [1]  2  4  6  8 10 12
```

```r
rep(c(1,0), 5)
```

```
#  [1] 1 0 1 0 1 0 1 0 1 0
```

Ex 7: Sorting and Ranking

Problem

Let `x <- c(40,15,30,10,50,25)`.

- Sort ascending and descending
- Find `order(x)` and verify `x[order(x)] == sort(x)`
- What rank does 30 receive?
- Sort `c("banana","apple","cherry")` alphabetically

Sol 7

```r
x <- c(40,15,30,10,50,25)
sort(x); sort(x, decreasing=TRUE)
```

```
# [1] 10 15 25 30 40 50
# [1] 50 40 30 25 15 10
```

```r
ord <- order(x); all(x[ord] == sort(x))
```

```
# [1] TRUE
```

```r
rank(x)  # 30 -> rank 4
```

```
# [1] 5 2 4 1 6 3
```

```r
sort(c("banana","apple","cherry"))
```

```
# [1] "apple"  "banana" "cherry"
```

Ex 8: NA, NULL, NaN, Inf

Problem

- Given `x <- c(3, NA, -1, NaN, Inf, -Inf, 0)`, apply `is.na`, `is.nan`, `is.finite`, `is.infinite`. Why is `is.na(NaN)` TRUE?
- Remove non-finite values and compute mean
- Show `c(1, NULL, 3)` vs `c(1, NA, 3)` lengths
- Compute `NA^0`, `NA | TRUE`, `NA & FALSE`

Sol 8

```r
x <- c(3, NA, -1, NaN, Inf, -Inf, 0)
is.na(x); is.nan(x); is.finite(x)
```

```
# [1] FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE
# [1] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE
# [1]  TRUE FALSE  TRUE FALSE FALSE FALSE  TRUE
```

```r
mean(x[is.finite(x)])
```

```
# [1] 0.6666667
```

```r
length(c(1, NULL, 3)); length(c(1, NA, 3))        # 2 vs 3
```

```
# [1] 2
# [1] 3
```

```r
NA^0; NA | TRUE; NA & FALSE
```

```
# [1] 1
# [1] TRUE
# [1] FALSE
```

Ex 9: Type Coercion

Problem

- What does `c(TRUE, 1L, 2.5, "x")` produce? What type?
- Coerce `c("3","1.5","NA","7")` to numeric. What happens to `"NA"`?
- Predict `as.integer(c(3.9, -2.7, TRUE, "5"))`
- Why does `1 == "1"` return `TRUE`?

Sol 9

```r
x <- c(TRUE, 1L, 2.5, "x"); x; typeof(x)          # character
```

```
# [1] "TRUE" "1"    "2.5"  "x"
# [1] "character"
```

```r
as.numeric(c("3","1.5","NA","7"))                  # NA + warning
```

```
# [1] 3.0 1.5  NA 7.0
```

```r
as.integer(c(3.9, -2.7, TRUE, "5"))                # 3,-2,1,5
```

```
# [1]  3 -2 NA  5
```

```r
1 == "1"  # 1 coerced to "1", then string compare
```

```
# [1] TRUE
```

Ex 10: `seq` and `seq_along`

Problem

- Generate `1.0 2.5 4.0 5.5 7.0 8.5 10.0` using `seq`
- Show `1:length(x)` is unsafe when `x <- NULL`; fix with `seq_along`
- 6 equally-spaced values from 0 to 2π
- What does `seq_along(integer(0))` return?

Sol 10

```r
seq(1, 10, by=1.5)
```

```
# [1]  1.0  2.5  4.0  5.5  7.0  8.5 10.0
```

```r
x <- NULL; 1:length(x)                          # c(1,0) WRONG
```

```
# [1] 1 0
```

```r
seq_along(x)                                     # integer(0) SAFE
```

```
# integer(0)
```

```r
seq(0, 2*pi, length.out=6)
```

```
# [1] 0.000000 1.256637 2.513274 3.769911 5.026548 6.283185
```

```r
seq_along(integer(0))                            # integer(0)
```

```
# integer(0)
```

Ex 11: `rep` Patterns

Problem

Using `rep`, generate:

- `1 1 2 2 3 3 4 4 5 5`
- `1 2 3 1 2 3 1 2 3 1 2 3`
- `"A" "A" "A" "B" "B" "C"`
- Alternating `TRUE FALSE` of length 20

Sol 11

```r
rep(1:5, each=2)
```

```
#  [1] 1 1 2 2 3 3 4 4 5 5
```

```r
rep(1:3, times=4)
```

```
#  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```r
rep(c("A","B","C"), c(3,2,1))
```

```
# [1] "A" "A" "A" "B" "B" "C"
```

```r
rep(c(TRUE,FALSE), 10)
```

```
#  [1]  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE
# [13]  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE
```

Ex 12: `%in%` and `match`

Problem

`haystack <- c(3,1,4,1,5,9,2,6,5,3)`; `needles <- c(5,7,2)`.

- Which needles are present?
- First occurrence position of each
- Difference between `match("a", c("b","a","a"))` and `which(c("b","a","a") == "a")`
- Filter `mtcars` rows where `cyl %in% c(4,6)`

Sol 12

```r
haystack <- c(3,1,4,1,5,9,2,6,5,3); needles <- c(5,7,2)
needles %in% haystack
```

```
# [1]  TRUE FALSE  TRUE
```

```r
match(needles, haystack)                          # NA for 7
```

```
# [1]  5 NA  7
```

```r
match("a", c("b","a","a"))                        # 2 (first)
```

```
# [1] 2
```

```r
which(c("b","a","a") == "a")                      # c(2,3) (all)
```

```
# [1] 2 3
```

```r
nrow(mtcars[mtcars$cyl %in% c(4,6), ])
```

```
# [1] 18
```

Ex 13: Vector Arithmetic

Problem

- Compute $H_{100} = \sum_{k=1}^{100} \frac{1}{k}$ in one line
- Compute $\sum_{k=0}^{50} \frac{(-1)^k}{k!}$. Compare with `exp(-1)`
- For `x <- seq(0.1,2,by=0.1)`, compute $e^x \sin x - \ln x$, rounded to 4 dp
- Compute `diff(cumsum(1:10))` and explain

Sol 13

```r
sum(1 / 1:100)                                  # H_100
```

```
# [1] 5.187378
```

```r
k <- 0:50; sum((-1)^k / factorial(k)); exp(-1)
```

```
# [1] 0.3678794
# [1] 0.3678794
```

```r
x <- seq(0.1, 2, by=0.1)
round(exp(x)*sin(x) - log(x), 4)
```

```
#  [1] 2.4129 1.8521 1.6029 1.4972 1.4836 1.5397 1.6540 1.8196 2.0320 2.2874
# [11] 2.5820 2.9122 3.2732 3.6597 4.0650 4.4809 4.8977 5.3036 5.6850 6.0257
```

```r
diff(cumsum(1:10))  # recovers 2:10
```

```
# [1]  2  3  4  5  6  7  8  9 10
```

Ex 14: Attributes and `structure`

Problem

- Create `a <- 1:4`; attach attributes `"unit"="kg"` and `"source"="lab"`. Print
- Recreate using `structure()` in one call
- `x <- 1:3` vs `y <- c(1,2,3)`: are they `identical`? `all.equal`? Why?
- When to use `isTRUE(all.equal())` in an `if`?

Sol 14

```r
a <- 1:4; attr(a,"unit") <- "kg"; attr(a,"source") <- "lab"
attributes(a)
```

```
# $unit
# [1] "kg"
#
# $source
# [1] "lab"
```

```r
b <- structure(1:4, unit="kg", source="lab")
x <- 1:3; y <- c(1,2,3)
identical(x, y); all.equal(x, y)                    # F; T
```

```
# [1] FALSE
```

```
# [1] TRUE
```

```r
isTRUE(all.equal(0.1+0.2, 0.3))                    # TRUE
```

```
# [1] TRUE
```

Ex 15: Factors

Problem

- Create factor `f <- factor(c("low","med","high","med","low"))`. What are the levels?
- Create an ordered factor with levels `low < med < high`. Test `f > "low"`
- Use `table(f)` to count occurrences
- What does `as.integer(f)` return and why?

Sol 15

```r
f <- factor(c("low","med","high","med","low"))
levels(f)                                          # alphabetical
```

```
# [1] "high" "low"  "med"
```

```r
fo <- factor(f, levels=c("low","med","high"), ordered=TRUE)
fo > "low"
```

```
# [1] FALSE  TRUE  TRUE  TRUE FALSE
```

```r
table(f)
```

```
# f
# high  low  med
#    1    2    2
```

```r
as.integer(f)  # underlying integer codes
```

```
# [1] 2 3 1 3 2
```

Ex 16: Floating Point

Problem

- Is `0.1 + 0.2 == 0.3` TRUE or FALSE? Why?
- Use `all.equal` to compare correctly
- What is `trunc(-3.7)` vs `floor(-3.7)`?
- Demonstrate banker's rounding: `round(0.5)`, `round(1.5)`, `round(2.5)`

Sol 16

```r
0.1 + 0.2 == 0.3  # FALSE (floating point)
```

```
# [1] FALSE
```

```r
all.equal(0.1 + 0.2, 0.3)                          # TRUE
```

```
# [1] TRUE
```

```r
trunc(-3.7); floor(-3.7)  # -3 vs -4
```

```
# [1] -3
# [1] -4
```

```r
round(0.5); round(1.5); round(2.5)  # 0; 2; 2
```

```
# [1] 0
# [1] 2
# [1] 2
```

Ex 17: Appending and Inserting

Problem

- Start with `v <- 1:5`. Append `c(10,11)` to the end
- Insert 99 after position 3 using `append`
- What happens if you assign `v[10] <- 42` when `length(v)` is 7?
- Remove the 3rd element without using negative indexing (use logical)

Sol 17

```r
v <- 1:5; v <- c(v, c(10,11)); v
```

```
# [1]  1  2  3  4  5 10 11
```

```r
append(v, 99, after=3)
```

```
# [1]  1  2  3 99  4  5 10 11
```

```r
v[10] <- 42; v   # NAs fill gaps
```

```
#  [1]  1  2  3  4  5 10 11 NA NA 42
```

```r
v[seq_along(v) != 3]
```

```
# [1]  1  2  4  5 10 11 NA NA 42
```

Ex 18: `sample` and `sequence`

Problem

- Simulate 20 dice rolls with `sample`
- Use `sample` to shuffle `letters[1:10]`
- Using `sequence`, generate `1 2 3 4  1 2 3  1 2  1`
- Generate `5 4 3 2 1  6 5 4 3 2 1` using `sequence`

Sol 18

```r
set.seed(42)
sample(1:6, 20, replace=TRUE)
```

```
#  [1] 1 5 1 1 2 4 2 2 1 4 1 5 6 4 2 2 3 1 1 3
```

```r
sample(letters[1:10])
```

```
#  [1] "d" "e" "g" "i" "h" "j" "b" "c" "f" "a"
```

```r
sequence(4:1)
```

```
#  [1] 1 2 3 4 1 2 3 1 2 1
```

```r
sequence(5:6, from=5:6, by=-1)
```

```
#  [1] 5 4 3 2 1 6 5 4 3 2 1
```

Operators and Common Functions (Ex 19–26)

Ex 19: Operator Precedence

Problem

- Without running, predict: `2 + 3 * 4`, `2^3^2`, `1:3 + 1`, `1:(3+1)`
- Why does `-2^2` give `-4` instead of `4`?
- Predict: `TRUE | FALSE & FALSE`
- Write `x <- 5; x %% 3 == 2 & x > 3` and trace operator precedence

Sol 19

```r
2 + 3 * 4      # 14: * before +
```

```
# [1] 14
```

```r
2^3^2           # 512: ^ right-associative = 2^9
```

```
# [1] 512
```

```r
1:3 + 1         # 2 3 4: : before +
```

```
# [1] 2 3 4
```

```r
1:(3+1)         # 1 2 3 4
```

```
# [1] 1 2 3 4
```

```r
-2^2            # -4: ^ before unary -
```

```
# [1] -4
```

```r
TRUE | FALSE & FALSE  # TRUE: & before |
```

```
# [1] TRUE
```

```r
x <- 5; x %% 3 == 2 & x > 3  # TRUE
```

```
# [1] TRUE
```

Ex 20: Math Functions

Problem

- Compute `floor`, `ceiling`, `trunc`, `round` of `-5.7`
- Compute $\binom{10}{3}$ and verify with `factorial`
- Compute `119 %/% 13` and `119 %% 13`. Verify `119 == 13 * (119 %/% 13) + 119 %% 13`
- `signif(12345678, 4)`

Sol 20

```r
floor(-5.7); ceiling(-5.7); trunc(-5.7); round(-5.7)
```

```
# [1] -6
# [1] -5
# [1] -5
# [1] -6
```

```r
choose(10, 3); factorial(10)/(factorial(3)*factorial(7))
```

```
# [1] 120
# [1] 120
```

```r
119 %/% 13; 119 %% 13
```

```
# [1] 9
# [1] 2
```

```r
119 == 13 * (119 %/% 13) + 119 %% 13              # TRUE
```

```
# [1] TRUE
```

```r
signif(12345678, 4)
```

Ex 21: Statistical Summary

Problem

`v <- c(4, 7, 2, 9, 1, 5, 8, 3, 6, 10)`.

- mean, median, var, sd, range
- `quantile(v, c(0.25, 0.75))` and IQR
- Compute z-scores: $(v - \bar{v})/s$. How many exceed $|z| > 1$?
- Use `diff(v)` to find the largest jump

Sol 21

```r
v <- c(4, 7, 2, 9, 1, 5, 8, 3, 6, 10)
mean(v); median(v); var(v); sd(v); range(v)
```

```
# [1] 5.5
# [1] 5.5
# [1] 9.166667
# [1] 3.02765
# [1]  1 10
```

```r
quantile(v, c(0.25, 0.75)); IQR(v)
```

```
#  25%  75%
# 3.25 7.75
# [1] 4.5
```

```r
z <- (v - mean(v)) / sd(v); sum(abs(z) > 1)
```

```
# [1] 4
```

```r
max(abs(diff(v)))                                  # largest jump
```

```
# [1] 8
```

Ex 22: `table` and `prop.table`

Problem

`x <- c("A","B","A","C","B","A","C","C","B","A")`.

- Frequency table of `x`
- Proportion table
- Create `y <- c(1,1,2,1,2,2,1,2,1,2)`. Cross-tabulate `x` and `y`; add margins
- Which category of `x` is most frequent?

Sol 22

```r
x <- c("A","B","A","C","B","A","C","C","B","A")
table(x)
```

```
# x
# A B C
# 4 3 3
```

```r
prop.table(table(x))
```

```
# x
#   A   B   C
# 0.4 0.3 0.3
```

```r
y <- c(1,1,2,1,2,2,1,2,1,2)
addmargins(table(x, y))
```

```
#       y
# x       1  2 Sum
#   A     1  3   4
#   B     2  1   3
#   C     2  1   3
#   Sum   5  5  10
```

Ex 23: Harmonic Series Convergence

Problem

- Compute $H_n = \sum_{k=1}^{n} \frac{1}{k}$ for $n \in \{10, 100, 1000, 10000\}$
- Compare each H_n with $\ln(n) + \gamma$ where $\gamma \approx 0.5772$
- Plot $H_n - \ln n$ for $n = 1, \ldots, 200$

Sol 23

```r
ns <- c(10, 100, 1000, 10000)
Hn <- sapply(ns, function(n) sum(1/1:n))
data.frac(n=ns, Hn=round(Hn,4),
          approx=round(log(ns)+0.5772,4))
```

```
#       n     Hn approx
# 1    10 2.9290 2.8798
# 2   100 5.1874 5.1824
# 3  1000 7.4855 7.4850
# 4 10000 9.7876 9.7875
```

```r
x <- 1:200; y <- cumsum(1/x) - log(x)
plot(x, y, type="l", ylab=expression(H[n]-ln(n)))
abline(h=0.5772, col="red", lty=2)
```

Ex 24: `pmin`/`pmax` Application

Problem

- Clip vector `v <- c(-5,3,12,-2,8,25)` to range $[0, 10]$ using `pmin` and `pmax`
- Given two portfolios `a <- c(5,8,3,9)` and `b <- c(7,2,6,4)`, compute the "best of both" portfolio
- Compute $\sum_{k=1}^{100} \min(2^k, k^4)$ without loops

Sol 24

```r
v <- c(-5, 3, 12, -2, 8, 25)
pmin(pmax(v, 0), 10)                                  # clip to [0,10]
```

```
# [1]  0  3 10  0  8 10
```

```r
a <- c(5,8,3,9); b <- c(7,2,6,4)
pmax(a, b)
```

```
# [1] 7 8 6 9
```

```r
k <- 1:100; sum(pmin(2^k, k^4))
```

```
# [1] 2050220551
```

Ex 25: Cumulative Functions

Problem

`x <- c(3, -1, 4, -1, 5, -9, 2, 6)`.

- Compute `cumsum(x)`. At which index does it first become negative?
- Maximum drawdown: $\max_i(\text{cummax}_i - \text{cumsum}_i)$
- Compute `cumprod(1 + c(0.05, -0.02, 0.03, -0.01))` (portfolio growth)

Sol 25

```r
x <- c(3, -1, 4, -1, 5, -9, 2, 6)
cs <- cumsum(x); cs
```

```
# [1]  3  2  6  5 10  1  3  9
```

```r
which(cs < 0)[1]                                  # first negative
```

```
# [1] NA
```

```r
cm <- cummax(cs); max(cm - cs)                    # max drawdown
```

```
# [1] 9
```

```r
cumprod(1 + c(0.05, -0.02, 0.03, -0.01))          # growth
```

```
# [1] 1.050000 1.029000 1.059870 1.049271
```

Ex 26: `order` and Data Sorting

Problem

`d <- data.frame(name=c("Zara","Amy","Mike","Bob"), score=c(88,95,72,95), age=c(22,20,25,21))`.

- Sort by score ascending
- Sort by score descending, ties broken by age ascending
- Find the name of the person with the highest score (youngest if tied)

Sol 26

```r
d <- data.frame(name=c("Zara","Amy","Mike","Bob"),
                score=c(88,95,72,95), age=c(22,20,25,21))
d[order(d$score), ]
```

```
#   name score age
# 3 Mike    72  25
# 1 Zara    88  22
# 2  Amy    95  20
# 4  Bob    95  21
```

```r
d[order(-d$score, d$age), ]
```

```
#   name score age
# 2  Amy    95  20
# 4  Bob    95  21
# 1 Zara    88  22
# 3 Mike    72  25
```

```r
d[order(-d$score, d$age), ]$name[1]                # Bob
```

```
# [1] "Amy"
```

Data Frames (Ex 27–40)

Ex 27: Creating and Inspecting

Problem

- Create a data frame with columns `name` (Alice, Bob, Cathy), `age` (25, 30, 28), `score` (88.5, 92.3, 76.8)
- Show `str`, `dim`, `names`, `summary`
- What class is `df$name`?
- Access `score` of row 2 by name and by position

Sol 27

```r
df <- data.frame(name=c("Alice","Bob","Cathy"),
                 age=c(25,30,28), score=c(88.5,92.3,76.8))
str(df); dim(df); names(df)
```

```
# 'data.frame': 3 obs. of  3 variables:
#  $ name : chr  "Alice" "Bob" "Cathy"
#  $ age  : num  25 30 28
#  $ score: num  88.5 92.3 76.8
```

```
# [1] 3 3
```

```
# [1] "name"  "age"   "score"
```

```r
class(df$name)
```

```
# [1] "character"
```

```r
df$score[2]; df[2, "score"]
```

```
# [1] 92.3
```

```
# [1] 92.3
```

Ex 28: Row Filtering

Problem

Using `tg <- ToothGrowth`:

- Rows where `supp == "OJ"` and `dose >= 1`
- Mean tooth length for VC vs OJ
- Rows where `len` is above its overall median
- Same as (a) but using `subset()`

Sol 28

```r
tg <- ToothGrowth
head(tg[tg$supp == "OJ" & tg$dose >= 1, ], 4)
```

```
#     len supp dose
# 41 19.7   OJ    1
# 42 23.3   OJ    1
# 43 23.6   OJ    1
# 44 26.4   OJ    1
```

```r
tapply(tg$len, tg$supp, mean)
```

```
#       OJ       VC
# 20.66333 16.96333
```

```r
head(tg[tg$len > median(tg$len), ], 4)
```

```
#     len supp dose
# 15 22.5   VC    1
# 21 23.6   VC    2
# 23 33.9   VC    2
# 24 25.5   VC    2
```

```r
head(subset(tg, supp=="OJ" & dose >= 1), 4)
```

Ex 29: Adding and Removing Columns

Problem

Start with `df <- data.frame(x=1:5, y=c(10,20,30,40,50))`.

- Add column `z = x * y`
- Add column `label` using `paste0("item", 1:5)`
- Remove column `z`
- Rename `y` to `value`

Sol 29

```r
df <- data.frame(x=1:5, y=c(10,20,30,40,50))
df$z <- df$x * df$y
df$label <- paste0("item", 1:5)
df$z <- NULL
names(df)[names(df)=="y"] <- "value"; df
```

```
#   x value label
# 1 1    10 item1
# 2 2    20 item2
# 3 3    30 item3
# 4 4    40 item4
# 5 5    50 item5
```

Ex 30: `with()` and `within()`

Problem

Using `tg <- ToothGrowth`:

- Use `with()` to compute `mean(len[dose == 2])`
- Use `within()` to add `log_len <- log(len)` and `len_centered <- len - mean(len)`

Sol 30

```r
tg <- ToothGrowth
with(tg, mean(len[dose == 2]))
```

```
# [1] 26.1
```

```r
tg2 <- within(tg, {
  log_len <- log(len)
  len_centered <- len - mean(len)
})
head(tg2, 3)
```

```
#    len supp dose len_centered  log_len
# 1  4.2   VC  0.5   -14.613333 1.435085
# 2 11.5   VC  0.5    -7.313333 2.442347
# 3  7.3   VC  0.5   -11.513333 1.987874
```

Ex 31: Handling NAs

Problem

`df <- data.frame(x=c(1,NA,3,4,NA), y=c(NA,2,3,NA,5), z=c(1,2,3,4,5))`.

- Which rows are complete?
- Remove all incomplete rows
- Replace NAs in column `x` with the mean of `x` (ignoring NAs)
- Count NAs per column

Sol 31

```r
df <- data.frame(x=c(1,NA,3,4,NA), y=c(NA,2,3,NA,5), z=1:5)
complete.cases(df)
```

```
# [1] FALSE FALSE  TRUE FALSE FALSE
```

```r
df[complete.cases(df), ]
```

```
#   x y z
# 3 3 3 3
```

```r
df$x[is.na(df$x)] <- mean(df$x, na.rm=TRUE); df
```

```
#          x  y z
# 1 1.000000 NA 1
# 2 2.666667  2 2
# 3 3.000000  3 3
# 4 4.000000 NA 4
# 5 2.666667  5 5
```

```r
colSums(is.na(df))
```

```
# x y z
# 0 2 0
```

Ex 32: `merge()`

Problem

`students <- data.frame(id=1:4, name=c("A","B","C","D")). grades <- data.frame(sid=c(1,2,2,4), course=c("Math","Math","Eng","Eng"), grade=c(85,90,78,92)).`

- Inner join on id
- Left join (keep all students)
- How many rows in each result? Why?

Sol 32

```r
students <- data.frame(id=1:4, name=c("A","B","C","D"))
grades <- data.frame(sid=c(1,2,2,4),
  course=c("Math","Math","Eng","Eng"), grade=c(85,90,78,92))
merge(students, grades, by.x="id", by.y="sid")     # inner
```

```
#   id name course grade
# 1  1    A   Math    85
# 2  2    B   Math    90
# 3  2    B    Eng    78
# 4  4    D    Eng    92
```

```r
merge(students, grades, by.x="id", by.y="sid",
      all.x=TRUE)                                  # left
```

```
#   id name course grade
# 1  1    A   Math    85
# 2  2    B   Math    90
# 3  2    B    Eng    78
# 4  3    C   <NA>    NA
# 5  4    D    Eng    92
```

```r
# inner: 4 rows; left: 5 rows (C has NA)
```

Ex 33: `aggregate()`

Problem

Using `mtcars`:

- Mean `mpg` by `cyl`
- Mean and sd of `mpg` by `cyl` and `am`
- Max `hp` by `cyl`

Sol 33

```r
aggregate(mpg ~ cyl, data=mtcars, FUN=mean)
```

```
#   cyl      mpg
# 1   4 26.66364
# 2   6 19.74286
# 3   8 15.10000
```

```r
aggregate(mpg ~ cyl + am, data=mtcars,
  FUN=function(x) c(m=mean(x), s=sd(x)))
```

```
#   cyl am      mpg.m      mpg.s
# 1   4  0 22.9000000  1.4525839
# 2   6  0 19.1250000  1.6317169
# 3   8  0 15.0500000  2.7743959
# 4   4  1 28.0750000  4.4838599
# 5   6  1 20.5666667  0.7505553
# 6   8  1 15.4000000  0.5656854
```

```r
aggregate(hp ~ cyl, data=mtcars, FUN=max)
```

```
#   cyl  hp
# 1   4 113
# 2   6 175
```

Ex 34: `cbind` and `rbind`

Problem

- Create two 1-column data frames and `cbind` them
- Create two data frames with same columns and `rbind`
- Use `do.call(rbind, list_of_dfs)` with 3 data frames

Sol 34

```r
d1 <- data.frame(a=1:3); d2 <- data.frame(b=4:6)
cbind(d1, d2)
```

```
#   a b
# 1 1 4
# 2 2 5
# 3 3 6
```

```r
e1 <- data.frame(x="a", y=1); e2 <- data.frame(x="b", y=2)
rbind(e1, e2)
```

```
#   x y
# 1 a 1
# 2 b 2
```

```r
dfs <- list(data.frame(x=1,y=2), data.frame(x=3,y=4),
            data.frame(x=5,y=6))
do.call(rbind, dfs)
```

```
#   x y
# 1 1 2
# 2 3 4
# 3 5 6
```

Ex 35: Data Frame as Named List

Problem

- Show `is.list(mtcars)` is TRUE. What does `typeof(mtcars)` return?
- Access the `mpg` column using `[[` notation
- What is the difference between `mtcars["mpg"]` and `mtcars[["mpg"]]`?
- Convert `mtcars[1:3, 1:3]` to a list and back

Sol 35

```r
is.list(mtcars); typeof(mtcars)                    # TRUE; "list"
```

```
# [1] TRUE
# [1] "list"
```

```r
head(mtcars[["mpg"]])
```

```
# [1] 21.0 21.0 22.8 21.4 18.7 18.1
```

```r
class(mtcars["mpg"]); class(mtcars[["mpg"]])       # df vs numeric
```

```
# [1] "data.frame"
# [1] "numeric"
```

```r
l <- as.list(mtcars[1:3, 1:3]); as.data.frame(l)
```

```
#    mpg cyl disp
# 1 21.0   6  160
# 2 21.0   6  160
# 3 22.8   4  108
```

Ex 36: `split()` on Data Frames

Problem

- Split `iris` by `Species`. What type is the result?
- Compute mean `Sepal.Length` per species using `split` + `sapply`
- Recombine with `do.call(rbind, ...)`

Sol 36

```r
sp <- split(iris, iris$Species)
class(sp); names(sp)                                # list
```

```
# [1] "list"
```

```
# [1] "setosa"     "versicolor" "virginica"
```

```r
sapply(sp, function(d) mean(d$Sepal.Length))
```

```
#     setosa versicolor  virginica
#      5.006      5.936      6.588
```

```r
head(do.call(rbind, sp), 4)
```

```
#          Sepal.Length Sepal.Width Petal.Length Petal.Width Species
# setosa.1          5.1         3.5          1.4         0.2  setosa
# setosa.2          4.9         3.0          1.4         0.2  setosa
# setosa.3          4.7         3.2          1.3         0.2  setosa
# setosa.4          4.6         3.1          1.5         0.2  setosa
```

Ex 37: `subset()` with `select`

Problem

Using `mtcars`:

- Select only `mpg`, `cyl`, `hp` for cars with `mpg > 25`
- Exclude `drat` and `wt` columns
- Why can you write `subset(mtcars, mpg > 25)` without quoting `mpg`?

Sol 37

```r
subset(mtcars, mpg > 25, select=c(mpg, cyl, hp))
```

```
#                 mpg cyl  hp
# Fiat 128       32.4   4  66
# Honda Civic    30.4   4  52
# Toyota Corolla 33.9   4  65
# Fiat X1-9      27.3   4  66
# Porsche 914-2  26.0   4  91
# Lotus Europa   30.4   4 113
```

```r
head(subset(mtcars, select=-c(drat, wt)), 3)
```

```
#                mpg cyl disp  hp  qsec vs am gear carb
# Mazda RX4     21.0   6  160 110 16.46  0  1    4    4
# Mazda RX4 Wag 21.0   6  160 110 17.02  0  1    4    4
# Datsun 710    22.8   4  108  93 18.61  1  1    4    1
```

```r
# subset uses non-standard evaluation: it captures
# the expression and evaluates it within the data frame
```

Ex 38: Sorting with Multiple Keys

Problem

`df <- data.frame(dept=c("A","B","A","B","A"), name=c("Zoe","Amy","Bob","Cal","Amy"), salary=c(50,60,70,55,65))`.

- Sort by dept ascending, then salary descending
- Find the highest-paid person in each department (using `order` + `!duplicated`)

Sol 38

```r
df <- data.frame(dept=c("A","B","A","B","A"),
  name=c("Zoe","Amy","Bob","Cal","Amy"), salary=c(50,60,70,55,65))
df[order(df$dept, -df$salary), ]
```

```
#   dept name salary
# 3    A  Bob     70
# 5    A  Amy     65
# 1    A  Zoe     50
# 2    B  Amy     60
# 4    B  Cal     55
```

```r
sorted <- df[order(df$dept, -df$salary), ]
sorted[!duplicated(sorted$dept), ]
```

```
#   dept name salary
# 3    A  Bob     70
# 2    B  Amy     60
```

Ex 39: Column-wise Operations

Problem

- Use `colMeans` on `mtcars[,1:4]`
- Use `sapply` to find `max` of each column
- Standardize (z-score) all numeric columns using `scale()`. Compare with manual `sweep`

Sol 39

```r
colMeans(mtcars[,1:4])
```

```
#       mpg       cyl      disp        hp
#  20.09062   6.18750 230.72188 146.68750
```

```r
sapply(mtcars[,1:4], max)
```

```
#   mpg   cyl  disp    hp
#  33.9   8.0 472.0 335.0
```

```r
head(scale(mtcars[,1:2]), 3)
```

```
#                     mpg        cyl
# Mazda RX4     0.1508848 -0.1049878
# Mazda RX4 Wag 0.1508848 -0.1049878
# Datsun 710    0.4495434 -1.2248578
```

```r
m <- as.matrix(mtcars[,1:2])
head(sweep(sweep(m, 2, colMeans(m)), 2, apply(m,2,sd), "/"), 3)
```

```
#                     mpg        cyl
# Mazda RX4     0.1508848 -0.1049878
# Mazda RX4 Wag 0.1508848 -0.1049878
# Datsun 710    0.4495434 -1.2248578
```

Ex 40: Building a Gradebook

Problem

- Create `grades <- data.frame(name=c("A","B","C","D","E"), hw=c(85,92,78,95,88), exam=c(90,85,92,88,76))`
- Add `final = 0.4*hw + 0.6*exam`
- Add `letter` using nested `ifelse` (A/B/C/F)
- Sort by `final` descending

Sol 40

```r
grades <- data.frame(name=c("A","B","C","D","E"),
  hw=c(85,92,78,95,88), exam=c(90,85,92,88,76))
grades$final <- 0.4*grades$hw + 0.6*grades$exam
grades$letter <- ifelse(grades$final>=90,"A",
  ifelse(grades$final>=80,"B",
  ifelse(grades$final>=70,"C","F")))
grades[order(-grades$final), ]
```

```
#   name hw exam final letter
# 4    D 95   88  90.8      A
# 1    A 85   90  88.0      B
# 2    B 92   85  87.8      B
# 3    C 78   92  86.4      B
# 5    E 88   76  80.8      B
```

Lists (Ex 41–52)

Ex 41: Creating and Accessing Lists

Problem

- Create `l <- list(name="Alice", scores=c(85,92,78), passed=TRUE)`
- Access `scores` with `$`, `[[]]`, and `[]`. Show the different classes
- Add element `grade = "B"`. Remove `passed`
- What is `length(l)` before and after?

Sol 41

```r
l <- list(name="Alice", scores=c(85,92,78), passed=TRUE)
l$scores; l[["scores"]]; l["scores"]
```

```
# [1] 85 92 78
# [1] 85 92 78
# $scores
# [1] 85 92 78
```

```r
class(l$scores); class(l["scores"])                # vector vs list
```

```
# [1] "numeric"
# [1] "list"
```

```r
l$grade <- "B"; l$passed <- NULL
length(l)  # now 3
```

```
# [1] 3
```

Ex 42: `unlist()` Behavior

Problem

- `l <- list(a=1:3, b=4:5)`. What does `unlist(l)` produce? Explain the names
- `unlist(l, use.names=FALSE)`
- `l2 <- list(x="hello", y=1:3)`. What type is `unlist(l2)`? Why?
- Mean of a list of numbers: `list(10, 20, 30)`

Sol 42

```r
l <- list(a=1:3, b=4:5)
unlist(l)                # named: a1 a2 a3 b1 b2
```

```
# a1 a2 a3 b1 b2
#  1  2  3  4  5
```

```r
unlist(l, use.names=FALSE)
```

```
# [1] 1 2 3 4 5
```

```r
unlist(list(x="hello", y=1:3))  # all character
```

```
#       x      y1      y2      y3
# "hello"     "1"     "2"     "3"
```

```r
mean(unlist(list(10, 20, 30)))                    # 20
```

```
# [1] 20
```

Ex 43: Nested Lists

Problem

- Create `student <- list(info=list(name="Bob", age=20), grades=list(math=90, eng=85))`
- Access Bob's math grade
- Add `grades$sci = 92`
- Use `str()` to visualize the structure

Sol 43

```r
student <- list(info=list(name="Bob", age=20),
                grades=list(math=90, eng=85))
student$grades$math                                  # 90
```

```
# [1] 90
```

```r
student$grades$sci <- 92
str(student)
```

```
# List of 2
#  $ info  :List of 2
#   ..$ name: chr "Bob"
#   ..$ age : num 20
#  $ grades:List of 3
#   ..$ math: num 90
#   ..$ eng : num 85
#   ..$ sci : num 92
```

Ex 44: `split()` and Recombine

Problem

- `split(mtcars$mpg, mtcars$cyl)` — what type?
- Compute `mean` per group using `sapply`
- Same using `tapply` in one line
- `split(1:10, rep(c("odd","even"), 5))`

Sol 44

```r
sp <- split(mtcars$mpg, mtcars$cyl); class(sp)    # list
```

```
# [1] "list"
```

```r
sapply(sp, mean)
```

```
#        4        6        8
# 26.66364 19.74286 15.10000
```

```r
tapply(mtcars$mpg, mtcars$cyl, mean)
```

```
#        4        6        8
# 26.66364 19.74286 15.10000
```

```r
split(1:10, rep(c("odd","even"), 5))
```

```
# $even
# [1]  2  4  6  8 10
#
# $odd
# [1] 1 3 5 7 9
```

Ex 45: List Concatenation

Problem

- Concatenate `list(a=1)` and `list(b=2)` with `c()`
- What happens with `c(list(a=1), 5)`?
- Build a list incrementally in a loop: for i=1:5, append `i^2`

Sol 45

```r
c(list(a=1), list(b=2))
```

```
# $a
# [1] 1
#
# $b
# [1] 2
```

```r
c(list(a=1), 5)                                    # 5 becomes element
```

```
# $a
# [1] 1
#
# [[2]]
# [1] 5
```

```r
l <- list()
for (i in 1:5) l[[i]] <- i^2; l
```

```
# [[1]]
# [1] 1
#
# [[2]]
```

Ex 46: Conversion Table

Problem

- Vector to list: `as.list(1:3)` vs `list(1:3)`. Explain difference
- List to data frame: `as.data.frame(list(a=1:3, b=4:6))`
- Data frame to matrix: what happens with `as.matrix(data.frame(x=1:3, y=c("a","b","c")))`?
- List to vector: why use `unlist` not `as.vector`?

Sol 46

```r
as.list(1:3)     # 3 elements
```

```
# [[1]]
# [1] 1
#
# [[2]]
# [1] 2
#
# [[3]]
# [1] 3
```

```r
list(1:3)        # 1 element (the vector)
```

```
# [[1]]
# [1] 1 2 3
```

```r
as.data.frame(list(a=1:3, b=4:6))
```

```
#   a b
# 1 1 4
# 2 2 5
# 3 3 6
```

Ex 47: Named List as Lookup Table

Problem

- Create a lookup: `grade_points <- list(A=4.0, B=3.0, C=2.0, D=1.0, F=0)`
- Look up GPA for grades `c("A","B","F","A","C")` using `sapply` + `[[`
- Compute mean GPA

Sol 47

```r
grade_points <- list(A=4.0, B=3.0, C=2.0, D=1.0, F=0)
g <- c("A","B","F","A","C")
gpa <- sapply(g, function(x) grade_points[[x]])
gpa
```

```
# A B F A C
# 4 3 0 4 2
```

```r
mean(gpa)                                    # 2.6
```

```
# [1] 2.6
```

Ex 48: `lapply/sapply` on Lists

Problem

`l <- list(a=1:10, b=rnorm(20), c=c(TRUE,FALSE,TRUE,TRUE)).`

- Compute `mean` of each element with `lapply` and `sapply`
- Compute `length` of each
- Apply custom function: range (max - min)

Sol 48

```r
set.seed(1)
l <- list(a=1:10, b=rnorm(20), c=c(TRUE,FALSE,TRUE,TRUE))
lapply(l, mean); sapply(l, mean)
```

```
# $a
# [1] 5.5
#
# $b
# [1] 0.1905239
#
# $c
# [1] 0.75
```

```
#         a         b         c
# 5.5000000 0.1905239 0.7500000
```

```r
sapply(l, length)
```

```
#  a  b  c
# 10 20  4
```

```r
sapply(l, function(x) diff(range(x)))
```

Ex 49: `Filter`, `Find`, `Position`

Problem

`l <- list(1:3, "hello", 4:6, TRUE, NULL, c(1.5, 2.5))`.

- Keep only numeric elements
- Find first character element
- Position of first character element
- Filter integers from `1:30` that are divisible by both 3 and 7

Sol 49

```r
l <- list(1:3, "hello", 4:6, TRUE, NULL, c(1.5, 2.5))
Filter(is.numeric, l)
```

```
# [[1]]
# [1] 1 2 3
#
# [[2]]
# [1] 4 5 6
#
# [[3]]
# [1] 1.5 2.5
```

```r
Find(is.character, l)                          # "hello"
```

```
# [1] "hello"
```

```r
Position(is.character, l)                      # 2
```

```
# [1] 2
```

```r
Filter(\(x) x%%3==0 & x%%7==0, 1:30)          # 21
```

```
# [1] 21
```

Ex 50: `Reduce()`

Problem

- Compute `1+2+3+4+5` using `Reduce("+", 1:5)`. Show accumulated steps
- Find intersection of 3 vectors via `Reduce(intersect, ...)`
- Merge 3 data frames pairwise using `Reduce`

Sol 50

```r
Reduce("+", 1:5)                                    # 15
```

```
# [1] 15
```

```r
Reduce("+", 1:5, accumulate=TRUE)                   # steps
```

```
# [1]  1  3  6 10 15
```

```r
vecs <- list(c(1,2,3,4), c(2,3,4,5), c(3,4,5,6))
Reduce(intersect, vecs)                             # 3 4
```

```
# [1] 3 4
```

```r
dfs <- list(data.frame(id=1:3, a=10:12),
            data.frame(id=2:4, b=20:22))
Reduce(\(x,y) merge(x,y,by="id",all=TRUE), dfs)
```

```
#   id  a  b
# 1  1 10 NA
# 2  2 11 20
# 3  3 12 21
# 4  4 NA 22
```

Ex 51: List Modification Patterns

Problem

- Given `l <- list(x=1, y=2, z=3)`, remove `x` and `z` in one operation
- Replace all numeric elements with their squares using `lapply`
- Deep copy a nested list and modify without affecting original

Sol 51

```r
l <- list(x=1, y=2, z=3)
l[c("x","z")] <- NULL; l                          # only y
```

```
# $y
# [1] 2
```

```r
l2 <- list(a=3, b="hi", c=4)
lapply(l2, function(x) if(is.numeric(x)) x^2 else x)
```

```
# $a
# [1] 9
#
# $b
# [1] "hi"
#
# $c
# [1] 16
```

```r
orig <- list(inner=list(val=10))
copy <- orig; copy$inner$val <- 99
orig$inner$val  # still 10 (R copies on modify)
```

```
# [1] 10
```

Ex 52: `rapply()` — Recursive Apply

Problem

`l <- list(a=1:3, b=list(c=4:6, d=list(e=7:9)))`.

- Apply `sum` to all leaf elements using `rapply(l, sum)`
- Use `rapply(l, sum, how="unlist")` vs `how="list"`
- Double all values with `rapply(l, function(x) x*2, how="replace")`

Sol 52

```r
l <- list(a=1:3, b=list(c=4:6, d=list(e=7:9)))
rapply(l, sum)
```

```
#     a   b.c b.d.e
#     6    15    24
```

```r
rapply(l, sum, how="unlist")                    # named vector
```

```
#     a   b.c b.d.e
#     6    15    24
```

```r
rapply(l, sum, how="list")                      # list structure
```

```
# $a
# [1] 6
#
# $b
# $b$c
# [1] 15
#
# $b$d
# $b$d$e
# [1] 24
```

Matrices (Ex 53–58)

Ex 53: Creating Matrices

Problem

- Create 3×4 matrix filled column-wise from `1:12`
- Create same matrix filled row-wise
- Name rows `r1,r2,r3` and columns `c1,...,c4`. Access `m["r2","c3"]`
- Create a 3×3 identity matrix with `diag`

Sol 53

```r
matrix(1:12, nrow=3)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    4    7   10
# [2,]    2    5    8   11
# [3,]    3    6    9   12
```

```r
matrix(1:12, nrow=3, byrow=TRUE)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    2    3    4
# [2,]    5    6    7    8
# [3,]    9   10   11   12
```

```r
m <- matrix(1:12, 3, 4)
rownames(m) <- paste0("r",1:3); colnames(m) <- paste0("c",1:4)
m["r2","c3"]                                          # 8
```

```
# [1] 8
```

```r
diag(3)
```

```
#      [,1] [,2] [,3]
# [1,]    1    0    0
```

Ex 54: Matrix Arithmetic

Problem

`A <- matrix(1:4, 2, 2); B <- matrix(5:8, 2, 2)`.

- Element-wise `*` vs matrix `%*%`
- Compute `t(A)`, `det(A)`, `solve(A)`
- Verify `A %*% solve(A)` is the identity
- `crossprod(A,B)` vs `t(A) %*% B`

Sol 54

```r
A <- matrix(1:4, 2, 2); B <- matrix(5:8, 2, 2)
A * B; A %*% B
```

```
#      [,1] [,2]
# [1,]    5   21
# [2,]   12   32
```

```
#      [,1] [,2]
# [1,]   23   31
# [2,]   34   46
```

```r
t(A); det(A); solve(A)
```

```
#      [,1] [,2]
# [1,]    1    2
# [2,]    3    4
```

```
# [1] -2
```

```
#      [,1] [,2]
# [1,]   -2  1.5
# [2,]    1 -0.5
```

```r
round(A %*% solve(A))                    # identity
```

Ex 55: `apply()` on Matrices

Problem

`m <- matrix(1:12, nrow=3)`.

- Row sums and column sums via `apply`
- Per-column range
- Standardize each column (subtract mean, divide by sd)
- Compare `apply(m,2,sum)` speed with `colSums(m)`

Sol 55

```r
m <- matrix(1:12, nrow=3)
apply(m, 1, sum); apply(m, 2, sum)
```

```
# [1] 22 26 30
```

```
# [1]  6 15 24 33
```

```r
apply(m, 2, range)
```

```
#      [,1] [,2] [,3] [,4]
# [1,]    1    4    7   10
# [2,]    3    6    9   12
```

```r
apply(m, 2, \(x) (x-mean(x))/sd(x))
```

```
#      [,1] [,2] [,3] [,4]
# [1,]   -1   -1   -1   -1
# [2,]    0    0    0    0
# [3,]    1    1    1    1
```

```r
all.equal(apply(m,2,sum), colSums(m))              # TRUE
```

```
# [1] TRUE
```

Ex 56: `outer()` Product

Problem

- Multiplication table 1..5 $\times$ 1..5 using `outer`
- Compute $f(x, y) = x^2 + y^2$ for `x=1:4`, `y=1:3`
- All pairwise `paste` of `c("a","b")` and `c("1","2","3")`

Sol 56

```r
outer(1:5, 1:5, "*")
```

```
#      [,1] [,2] [,3] [,4] [,5]
# [1,]    1    2    3    4    5
# [2,]    2    4    6    8   10
# [3,]    3    6    9   12   15
# [4,]    4    8   12   16   20
# [5,]    5   10   15   20   25
```

```r
outer(1:4, 1:3, \(x,y) x^2 + y^2)
```

```
#      [,1] [,2] [,3]
# [1,]    2    5   10
# [2,]    5    8   13
# [3,]   10   13   18
# [4,]   17   20   25
```

```r
outer(c("a","b"), c("1","2","3"), paste0)
```

```
#      [,1] [,2] [,3]
# [1,] "a1" "a2" "a3"
# [2,] "b1" "b2" "b3"
```

Ex 57: Matrix Row/Column Selection

Problem

`m <- matrix(1:12, 3, 4)`.

- Extract row 2 as a vector vs as a 1-row matrix (`drop`)
- Extract column 3 both ways
- Select submatrix: rows 1-2, columns 2-3
- Replace all elements > 8 with `NA`

Sol 57

```r
m <- matrix(1:12, 3, 4)
m[2,]; m[2,,drop=FALSE]
```

```
# [1]  2  5  8 11
#      [,1] [,2] [,3] [,4]
# [1,]    2    5    8   11
```

```r
m[,3]; m[,3,drop=FALSE]
```

```
# [1] 7 8 9
#      [,1]
# [1,]    7
# [2,]    8
# [3,]    9
```

```r
m[1:2, 2:3]
```

```
#      [,1] [,2]
# [1,]    4    7
# [2,]    5    8
```

```r
m[m > 8] <- NA; m
```

Ex 58: Linear System

Problem

Solve $2x + 3y = 7,\ 4x - y = 1$.

- Set up as `A %*% x = b` and solve with `solve(A, b)`
- Verify by multiplying back
- Compute eigenvalues of `A`

Sol 58

```r
A <- matrix(c(2,4,3,-1), 2, 2); b <- c(7,1)
x <- solve(A, b); x
```

```
# [1] 0.7142857 1.8571429
```

```r
A %*% x  # should be c(7,1)
```

```
#      [,1]
# [1,]    7
# [2,]    1
```

```r
eigen(A)$values
```

```
# [1]  4.274917 -3.274917
```

Programming (Ex 59–84)

Ex 59: Default Arguments

Problem

- Write `power(x, n=2)` that computes `x^n`. Test with 1 and 2 arguments
- Write `greet(name, greeting="Hello")` that returns `"Hello, Alice!"`
- What happens if you call `power(n=3)` without `x`?

Sol 59

```r
power <- function(x, n=2) x^n
power(3); power(3, 3); power(x=5, n=2)
```

```
# [1] 9
# [1] 27
# [1] 25
```

```r
greet <- function(name, greeting="Hello")
  paste0(greeting, ", ", name, "!")
greet("Alice"); greet("Bob", "Hi")
```

```
# [1] "Hello, Alice!"
# [1] "Hi, Bob!"
```

```r
power(n=3)  # error: x is missing
```

```
# Error in power(n = 3): argument "x" is missing, with no default
```

Ex 60: Functions as Arguments

Problem

- Write `apply_fn(x, fn)` that applies `fn` to `x`. Test with `sqrt`, `log`, `abs`
- Extend to `apply_fn(x, fn, ...)` supporting extra arguments. Test: `apply_fn(c(1,NA,3), mean, na.rm=TRUE)`

Sol 60

```r
apply_fn <- function(x, fn, ...) fn(x, ...)
apply_fn(c(4,9,16), sqrt)
```

```
# [1] 2 3 4
```

```r
apply_fn(c(1,NA,3), mean, na.rm=TRUE)             # 2
```

```
# [1] 2
```

Ex 61: Implicit Return

Problem

- Write two versions of average: one with explicit `return()`, one without
- Write `bad_avg` that assigns to a local variable as last line. Show it returns silently
- Capture the invisible result with `x <- bad_avg(1:10)`

Sol 61

```r
avg_explicit <- function(x) { return(sum(x)/length(x)) }
avg_implicit <- function(x) { sum(x)/length(x) }
avg_explicit(1:10); avg_implicit(1:10)             # same
```

```
# [1] 5.5
```

```
# [1] 5.5
```

```r
bad_avg <- function(x) { result <- sum(x)/length(x) }
bad_avg(1:10)  # no output!
x <- bad_avg(1:10); x  # but value is captured
```

```
# [1] 5.5
```

Ex 62: Anonymous Functions

Problem

- Use `sapply(1:5, function(x) x^2 + 1)`
- Rewrite with `\(x)` shorthand
- Use anonymous function in `Filter` to keep even numbers from `1:20`
- Nested: `sapply(1:3, \(i) sapply(1:3, \(j) i*j))`

Sol 62

```r
sapply(1:5, function(x) x^2+1)
```

```
# [1]  2  5 10 17 26
```

```r
sapply(1:5, \(x) x^2+1)
```

```
# [1]  2  5 10 17 26
```

```r
Filter(\(x) x%%2==0, 1:20)
```

```
#  [1]  2  4  6  8 10 12 14 16 18 20
```

```r
sapply(1:3, \(i) sapply(1:3, \(j) i*j))            # 3x3 matrix
```

```
#      [,1] [,2] [,3]
# [1,]    1    2    3
# [2,]    2    4    6
# [3,]    3    6    9
```

Ex 63: Closures

Problem

- Write `make_adder(n)` that returns a function adding `n`. Test `add5 <- make_adder(5); add5(3)`
- Write `make_counter()` returning a list with `increment`, `get`, `reset` functions using `<<-`
- Verify the counter works across multiple calls

Sol 63

```r
make_adder <- function(n) function(x) x + n
add5 <- make_adder(5); add5(3)                    # 8
```

```
# [1] 8
```

```r
make_counter <- function() {
  n <- 0
  list(inc = function() {n <<- n+1; n},
       get = function() n,
       reset = function() {n <<- 0})
}
ctr <- make_counter()
ctr$inc(); ctr$inc(); ctr$get()                   # 2
```

```
# [1] 1
# [1] 2
# [1] 2
```

```r
ctr$reset(); ctr$get()                            # 0
```

```
# [1] 0
```

Ex 64: Dot-Prefix Naming

Problem

- Define `.helper <- function(x) x+1` and `public <- function(x) .helper(x)*2`
- Show `.helper` doesn't appear in `ls()` but does in `ls(all.names=TRUE)`
- Name 3 built-in R objects starting with `.`

Sol 64

```r
e <- new.env()
e$.helper <- function(x) x+1
e$public  <- function(x) e$.helper(x)*2
e$public(5)  # 12
```

```
# [1] 12
```

```r
ls(e); ls(e, all.names=TRUE)
```

```
# [1] "public"
```

```
# [1] ".helper" "public"
```

```r
# .Machine, .Platform, .GlobalEnv
```

Ex 65: `if (FALSE)` Idiom

Problem

- Wrap code in `if (FALSE) \{ ... \}` and show it doesn't execute
- Change to `if (TRUE)` and show it runs
- Why is this better than commenting with # for multi-line blocks?

Sol 65

```r
if (FALSE) {
  cat("This won't print\n")
  x <- 999
}
exists("x")  # from previous chunks, may be TRUE
```

```
# [1] TRUE
```

```r
if (TRUE) { cat("This prints\n") }
```

```
# This prints
```

```r
# Unlike #, the code inside is still parsed by R,
# so syntax errors are caught immediately.
```

Ex 66: Local Helper Functions

Problem

- Write `outer_fn(x)` that defines inner helper `validate(v)` checking `is.numeric(v) && length(v) > 0`, then uses it
- Show that `validate` is not accessible outside `outer_fn`
- Why is this pattern useful?

Sol 66

```r
outer_fn <- function(x) {
  validate <- function(v) is.numeric(v) && length(v) > 0
  if (!validate(x)) stop("invalid input")
  mean(x)
}
outer_fn(1:10)                                    # 5.5
```

```
# [1] 5.5
```

```r
outer_fn("abc")                                   # error
```

```
# Error in outer_fn("abc"): invalid input
```

```r
# validate is local, not in global env
# keeps global namespace clean; helper has access to
#     outer function's variables via lexical scoping
```

Ex 67: `if/else` and `ifelse`

Problem

- Write scalar `classify(x)` returning "pos", "neg", or "zero" using `if/else`
- Vectorize: classify `c(-3,0,5,2,-1)` using `ifelse`
- Nested `ifelse` to map scores to grades: ≥ 90 A, ≥ 80 B, ≥ 70 C, else F

Sol 67

```r
classify <- function(x) {
  if (x > 0) "pos" else if (x < 0) "neg" else "zero"
}
classify(5); classify(-1); classify(0)
```

```
# [1] "pos"
# [1] "neg"
# [1] "zero"
```

```r
ifelse(c(-3,0,5,2,-1)>0, "pos",
  ifelse(c(-3,0,5,2,-1)<0, "neg", "zero"))
```

```
# [1] "neg"  "zero" "pos"  "pos"  "neg"
```

```r
s <- c(95,82,71,65,90)
ifelse(s>=90,"A",ifelse(s>=80,"B",ifelse(s>=70,"C","F")))
```

```
# [1] "A" "B" "C" "F" "A"
```

Ex 68: `switch()`

Problem

- Write `calc(op, a, b)` using `switch(op, "+"=a+b, "-"=a-b, "*"=a*b, "/"=a/b, stop("unknown"))`
- Test with all 4 operators and an invalid one
- Write `day_type(day)` classifying weekdays vs weekends

Sol 68

```r
calc <- function(op, a, b)
  switch(op, "+"=a+b, "-"=a-b, "*"=a*b, "/"=a/b,
         stop("unknown op"))
calc("+",3,4); calc("*",5,6); calc("/",10,3)
```

```
# [1] 7
# [1] 30
# [1] 3.333333
```

```r
calc("^",2,3)  # error
```

```
# Error in calc("^", 2, 3): unknown op
```

```r
day_type <- function(d) switch(d, Monday=,Tuesday=,
  Wednesday=,Thursday=,Friday="Weekday",
  Saturday=,Sunday="Weekend","Unknown")
day_type("Monday"); day_type("Sunday")
```

```
# [1] "Weekday"
# [1] "Weekend"
```

Ex 69: `&&`/`||` vs `&`/`|`

Problem

- Show `&&` short-circuits: `FALSE && stop("boom")` doesn't error
- Show `&` doesn't short-circuit: `c(FALSE,TRUE) & c(TRUE,TRUE)` is vectorized
- Write safe null-check: `if (!is.null(x) && x > 0)` and explain why `&&` is required

Sol 69

```r
FALSE && stop("boom")  # no error: short-circuit
```

```
# [1] FALSE
```

```r
c(FALSE,TRUE) & c(TRUE,TRUE)                        # vectorized
```

```
# [1] FALSE  TRUE
```

```r
x <- NULL
if (!is.null(x) && x > 0) cat("pos") else cat("safe")
```

```
# safe
```

```r
# && stops after !is.null(x) is FALSE; & would try x>0
```

Ex 70: `for` Loop — FizzBuzz

Problem

- Classic FizzBuzz 1-20: print "Fizz" if divisible by 3, "Buzz" if by 5, "FizzBuzz" if both, else the number
- Vectorized version using `ifelse` (no loop)
- Compare with `paste0` trick

Sol 70

```r
for (i in 1:20) {
  if (i%%15==0) cat("FizzBuzz ")
  else if (i%%3==0) cat("Fizz ")
  else if (i%%5==0) cat("Buzz ")
  else cat(i, "")
}
```

```
# 1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz 16 17 Fizz 19 Buzz
```

```r
cat("\n")
```

```r
x <- 1:20
ifelse(x%%15==0,"FizzBuzz",ifelse(x%%3==0,"Fizz",
  ifelse(x%%5==0,"Buzz",as.character(x))))
```

```
#  [1] "1"        "2"        "Fizz"     "4"        "Buzz"     "Fizz"
#  [7] "7"        "8"        "Fizz"     "Buzz"     "11"       "Fizz"
# [13] "13"       "14"       "FizzBuzz" "16"       "17"       "Fizz"
# [19] "19"       "Buzz"
```

Ex 71: `while` Loop — Newton's Method

Problem

Implement Newton's method for $\sqrt{a}$: $x_{n+1} = \frac{1}{2}(x_n + a/x_n)$.

- Write `my_sqrt(a, tol=1e-10)` using `while`
- Test on 2, 9, 100
- Count iterations needed for `a=2`

Sol 71

```r
my_sqrt <- function(a, tol=1e-10) {
  x <- a; n <- 0
  while (TRUE) {
    x_new <- 0.5 * (x + a/x); n <- n + 1
    if (abs(x_new - x) < tol) break
    x <- x_new
  }
  list(root=x_new, iterations=n)
}
my_sqrt(2); my_sqrt(9)$root; my_sqrt(100)$root
```

```
# $root
# [1] 1.414214
#
# $iterations
# [1] 5
# [1] 3
# [1] 10
```

```r
my_sqrt(2)$iterations
```

Ex 72: `repeat/break` — Collatz

Problem

Collatz: if n even, $n/2$; if odd, $3n+1$. Conjecture: always reaches 1.

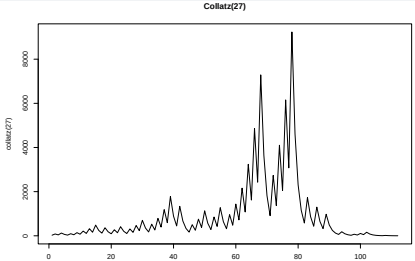
- Write `collatz(n)` returning the sequence using `repeat/break`
- Find the starting number 1–100 with the longest sequence
- Plot the sequence for `n=27`

Sol 72

```
collatz <- function(n) {
  seq <- n
  repeat { if (n==1) break; n <- if(n%%2==0) n/2 else 3*n+1
           seq <- c(seq, n) }
  seq
}
lengths <- sapply(1:100, \(n) length(collatz(n)))
which.max(lengths)  # starting number
```

```
# [1] 97
```

```
plot(collatz(27), type="l", main="Collatz(27)")
```



Ex 73: `next` — Filtered Sum

Problem

- Sum integers 1-100 that are NOT multiples of 7, using `for+next`
- Same with vectorized filtering (no loop)
- Sum only prime numbers 1-50. Write `is_prime(n)` helper

Sol 73

```r
s <- 0; for(i in 1:100) {if(i%%7==0) next; s<-s+i}; s
```

```
# [1] 4315
```

```r
sum((1:100)[(1:100)%%7 != 0])                    # same
```

```
# [1] 4315
```

```r
is_prime <- function(n) {
  if (n < 2) return(FALSE)
  if (n == 2) return(TRUE)
  all(n %% 2:floor(sqrt(n)) != 0)
}
sum(Filter(is_prime, 1:50))
```

```
# [1] 325
```

Ex 74: Vectorization Speed

Problem

- Write loop version of `sum(x^2)` for `x <- 1:1e5`
- Compare `system.time` with vectorized `sum(x^2)`
- Pre-allocate vs grow-in-loop: time `v <- c(); for(...) v <- c(v,i)` vs `v <- numeric(n); for(...) v[i] <- i`

Sol 74

```r
n <- 1e5; x <- 1:n
loop_sum <- function(x) {s<-0; for(v in x) s<-s+v^2; s}
system.time(loop_sum(x))                          # slow
```

```
#    user  system elapsed
#   0.003   0.000   0.004
```

```r
system.time(sum(x^2))                             # fast
```

```
#    user  system elapsed
#   0.001   0.000   0.001
```

```r
system.time({v<-c(); for(i in 1:n) v<-c(v,i)})    # very slow
```

```
#    user  system elapsed
#   4.637   0.098   4.735
```

```r
system.time({v<-numeric(n); for(i in 1:n) v[i]<-i}) # faster
```

```
#    user  system elapsed
#   0.005   0.001   0.004
```

Ex 75: Sieve of Eratosthenes

Problem

- Implement `sieve(n)` returning all primes up to `n` using a logical vector and nested loops
- Test with `sieve(50)`
- How many primes below 1000?

Sol 75

```r
sieve <- function(n) {
  is_p <- rep(TRUE, n); is_p[1] <- FALSE
  for (i in 2:floor(sqrt(n)))
    if (is_p[i]) is_p[seq(i*2, n, i)] <- FALSE
  which(is_p)
}
sieve(50)
```

```
#  [1]  2  3  5  7 11 13 17 19 23 29 31 37 41 43 47
```

```r
length(sieve(1000))                                  # 168
```

```
# [1] 168
```

Ex 76: Recursive Factorial and Fibonacci

Problem

- Write `my_fact(n)` using `Recall`
- Write `fib(n)` using `Recall`
- Compute `sapply(1:10, fib)`
- Why is naive recursive `fib` slow for large `n`?

Sol 76

```r
my_fact <- function(n) {
  if (n <= 1) return(1); n * Recall(n-1)
}
my_fact(5); my_fact(10)
```

```
# [1] 120
# [1] 3628800
```

```r
fib <- function(n) {
  if (n <= 2) return(1); Recall(n-1) + Recall(n-2)
}
sapply(1:10, fib)
```

```
#  [1]  1  1  2  3  5  8 13 21 34 55
```

```r
# exponential time: each call branches into 2
```

Ex 77: `match.arg` and `missing`

Problem

- Write `summary_stat(x, type=c("mean","median","sd"))` using `match.arg`. Test partial matching
- Write `f(x, y)` that checks `missing(y)` and uses default `y=0`
- Show `nargs()` inside a function

Sol 77

```r
summary_stat <- function(x, type=c("mean","median","sd")) {
  type <- match.arg(type)
  switch(type, mean=mean(x), median=median(x), sd=sd(x))
}
summary_stat(1:10, "mea")                          # mean
```

```
# [1] 5.5
```

```r
f <- function(x, y) {
  if (missing(y)) y <- 0; x + y
}
f(5); f(5, 3)
```

```
# [1] 5
# [1] 8
```

```r
g <- function(a,b,c) nargs(); g(1); g(1,2,3)
```

```
# [1] 1
# [1] 3
```

Ex 78: `invisible` and Side Effects

Problem

- Write `my_print(x)` that cats and returns `invisible(x)`. Show it doesn't print the return
- Capture the invisible return
- Write `log_and_return(x)` that prints a message and returns `invisible(NULL)`

Sol 78

```r
my_print <- function(x) { cat("Value:", x, "\n"); invisible(x) }
my_print(42)          # prints message, no [1] 42
```

```
# Value: 42
```

```r
val <- my_print(42); val                          # captured
```

```
# Value: 42
# [1] 42
```

```r
log_return <- function(x) {
  cat("Processed:", x, "\n"); invisible(NULL)
}
r <- log_return(5); is.null(r)                    # TRUE
```

```
# Processed: 5
# [1] TRUE
```

Ex 79: `on.exit` with `add=TRUE`

Problem

- Write function that sets `options(digits=3)`, uses `on.exit` to restore
- Write function with two `on.exit(add=TRUE)` calls. Show both run
- Show cleanup runs even on error

Sol 79

```r
show_digits <- function() {
  old <- options(digits=3)
  on.exit(options(old))
  cat("Inside:", getOption("digits"), "\n")
}
show_digits(); cat("Outside:", getOption("digits"), "\n")
```

```
# Inside: 3
# Outside: 7
```

```r
two_exits <- function() {
  on.exit(cat("exit1\n"), add=TRUE)
  on.exit(cat("exit2\n"), add=TRUE)
  cat("body\n")
}
two_exits()
```

```
# body
# exit1
# exit2
```

```r
tryCatch({
```

Ex 80: `Vectorize()`

Problem

- Write scalar `classify(x)` using `if/else` (not vectorized)
- Use `Vectorize` to create vector version. Test on `c(-2, 0, 5)`
- Why is `Vectorize` just a wrapper around `mapply`?

Sol 80

```r
classify <- function(x) {
  if (x > 0) "pos" else if (x < 0) "neg" else "zero"
}
classify_v <- Vectorize(classify)
classify_v(c(-2, 0, 5))
```

```
# [1] "neg"  "zero" "pos"
```

```r
# Vectorize calls mapply internally, applying the
# scalar function element-wise. Not truly vectorized (still loops).
```

Ex 81: Native Pipe |>

Problem

- Rewrite `round(mean(abs(c(-3,4,-1,2))), 2)` using `|>`
- Pipe `mtcars` through `subset` then `head`
- Use _ placeholder: `mtcars |> lm(mpg ~ wt, data = _)`

Sol 81

```r
c(-3,4,-1,2) |> abs() |> mean() |> round(2)
```

```
# [1] 2.5
```

```r
mtcars |> subset(cyl == 4) |> head(3)
```

```
#             mpg cyl  disp hp drat   wt  qsec vs am gear carb
# Datsun 710 22.8   4 108.0 93 3.85 2.32 18.61  1  1    4    1
# Merc 240D  24.4   4 146.7 62 3.69 3.19 20.00  1  0    4    2
# Merc 230   22.8   4 140.8 95 3.92 3.15 22.90  1  0    4    2
```

```r
mtcars |> lm(mpg ~ wt, data = _) |>
  coef()
```

```
# (Intercept)          wt
#   37.285126   -5.344472
```

Ex 82: `do.call`

Problem

- Use `do.call(paste, c(as.list(c("a","b","c")), sep="-"))`
- Use `do.call(rbind, list_of_dfs)` to stack 3 data frames
- Build a call to `seq` dynamically from a list of arguments

Sol 82

```r
do.call(paste, c(as.list(c("a","b","c")), sep="-"))
```

```
# [1] "a-b-c"
```

```r
dfs <- list(data.frame(x=1,y=2), data.frame(x=3,y=4))
do.call(rbind, dfs)
```

```
#   x y
# 1 1 2
# 2 3 4
```

```r
do.call(seq, list(from=1, to=10, by=2))
```

```
# [1] 1 3 5 7 9
```

Ex 83: `tryCatch` and `try`

Problem

- Use `tryCatch(log("a"), error=function(e) NA)`
- Loop over `list(1,"a",4)`, wrapping each in `try`. Keep only successes
- Write `safe_div(a,b)` that stops on `b==0` and test with `tryCatch`

Sol 83

```r
tryCatch(log("a"), error=function(e) NA)
```

```
# [1] NA
```

```r
inputs <- list(1, "a", 4)
results <- lapply(inputs, \(x) try(sqrt(x), silent=TRUE))
Filter(\(r) !inherits(r,"try-error"), results)
```

```
# [[1]]
# [1] 1
#
# [[2]]
# [1] 2
```

```r
safe_div <- function(a,b) {if(b==0) stop("div0!"); a/b}
tryCatch(safe_div(10,0), error=function(e) Inf)
```

```
# [1] Inf
```

Ex 84: stopifnot and warning

Problem

- Write `my_mean(x)` with `stopifnot(is.numeric(x), length(x)>0)`
- Write `safe_sqrt(x)` that issues `warning` for negative input and returns `sqrt(abs(x))`
- Use `suppressWarnings` to silence it

Sol 84

```r
my_mean <- function(x) {
  stopifnot(is.numeric(x), length(x) > 0)
  sum(x)/length(x)
}
my_mean(1:5); my_mean("abc")
```

```
# [1] 3
```

```
# Error in my_mean("abc"): is.numeric(x) is not TRUE
```

```r
safe_sqrt <- function(x) {
  if (any(x < 0)) warning("Negative values!")
  sqrt(abs(x))
}
safe_sqrt(c(4, -9, 16))
```

```
# [1] 2 3 4
```

```r
suppressWarnings(safe_sqrt(-1))
```

```
# [1] 1
```

Apply Family (Ex 85–96)

Ex 85: `apply` — Row and Column

Problem

`m <- matrix(c(3,1,4,1,5,9,2,6,5,3,5,8), nrow=3)`.

- Column means via `apply`
- Row ranges
- Apply custom function returning a named vector per column
- Apply on a data frame's numeric columns

Sol 85

```r
m <- matrix(c(3,1,4,1,5,9,2,6,5,3,5,8), nrow=3)
apply(m, 2, mean)
```

```
# [1] 2.666667 5.000000 4.333333 5.333333
```

```r
apply(m, 1, range)                                  # 2x3 matrix
```

```
#      [,1] [,2] [,3]
# [1,]    1    1    4
# [2,]    3    6    9
```

```r
apply(m, 2, \(x) c(mu=mean(x), sd=sd(x)))
```

```
#        [,1] [,2]     [,3]     [,4]
# mu 2.666667    5 4.333333 5.333333
# sd 1.527525    4 2.081666 2.516611
```

```r
apply(mtcars[1:3, 1:4], 2, mean)
```

```
#        mpg        cyl       disp         hp
#  21.600000   5.333333 142.666667 104.333333
```

Ex 86: `lapply` and `sapply`

Problem

`l <- list(a=1:10, b=seq(0,1,by=0.1), c=c(T,F,T,T,F))`.

- `sapply(l, length)`
- `sapply(l, range)` — what shape?
- `vapply(l, mean, numeric(1))`
- Why prefer `vapply` over `sapply`?

Sol 86

```r
l <- list(a=1:10, b=seq(0,1,by=0.1), c=c(T,F,T,T,F))
sapply(l, length)                                   # named vector
```

```
#  a  b  c
# 10 11  5
```

```r
sapply(l, range)                                    # 2x3 matrix
```

```
#       a b c
# [1,]  1 0 0
# [2,] 10 1 1
```

```r
vapply(l, mean, numeric(1))                         # named vector
```

```
#   a   b   c
# 5.5 0.5 0.6
```

```r
# vapply guarantees return type; sapply may return list
# unexpectedly if elements have different lengths
```

Ex 87: `tapply`

Problem

- Mean `mpg` by `cyl` in `mtcars`
- Two-way: mean `mpg` by `cyl` $\times$ `am`
- Use `tapply` on `ToothGrowth`: mean `len` by `supp` and `dose`

Sol 87

```r
tapply(mtcars$mpg, mtcars$cyl, mean)
```

```
#        4        6        8
# 26.66364 19.74286 15.10000
```

```r
tapply(mtcars$mpg, list(mtcars$cyl, mtcars$am), mean)
```

```
#        0        1
# 4 22.900 28.07500
# 6 19.125 20.56667
# 8 15.050 15.40000
```

```r
tapply(ToothGrowth$len,
  list(ToothGrowth$supp, ToothGrowth$dose), mean)
```

```
#      0.5     1     2
# OJ 13.23 22.70 26.06
# VC  7.98 16.77 26.14
```

Ex 88: `mapply` and `Map`

Problem

- `mapply(function(x,y) x+y, 1:5, 10:14)`
- `mapply(rep, 1:4, 4:1)`
- `Map("+", 1:3, 10:12)` — what type?
- Use `Map` to apply `paste` element-wise to two vectors

Sol 88

```r
mapply(function(x,y) x+y, 1:5, 10:14)
```

```
# [1] 11 13 15 17 19
```

```r
mapply(rep, 1:4, 4:1)                          # list
```

```
# [[1]]
# [1] 1 1 1 1
#
# [[2]]
# [1] 2 2 2
#
# [[3]]
# [1] 3 3
#
# [[4]]
# [1] 4
```

```r
Map("+", 1:3, 10:12)                           # list
```

```
# [[1]]
# [1] 11
#
```

Ex 89: `Reduce` Applications

Problem

- Factorial via `Reduce("*", 1:6)`
- Cumulative products via `accumulate=TRUE`
- Find the common elements of 4 vectors using `Reduce(intersect, ...)`

Sol 89

```r
Reduce("*", 1:6)                                    # 720
```

```
# [1] 720
```

```r
Reduce("*", 1:6, accumulate=TRUE)
```

```
# [1]   1   2   6  24 120 720
```

```r
Reduce(intersect, list(1:10, 3:12, 5:15, 7:20))   # 7:10
```

```
# [1]  7  8  9 10
```

Ex 90: Functional Pipeline

Problem

Process `x <- c(-5, 3, -1, 8, -2, 7)`:

- Filter positive → square → sum. Do with nested calls
- Same with pipe `|>`
- Same with `Reduce` composing 3 functions

Sol 90

```r
x <- c(-5, 3, -1, 8, -2, 7)
sum(x[x>0]^2)                                    # nested
```

```
# [1] 122
```

```r
x[x>0] |> (\(v) v^2)() |> sum()                   # pipe
```

```
# [1] 122
```

```r
fns <- list(\(v) v[v>0], \(v) v^2, sum)
Reduce(\(v,f) f(v), fns, init=x)
```

```
# [1] 122
```

Ex 91: `tapply` + `aggregate` Pipeline

Problem

Using `iris`:

- Mean `Sepal.Length` by `Species` with `tapply`
- Same with `aggregate`
- Mean of all numeric columns by `Species` with `aggregate(. ~ Species, ...)`

Sol 91

```r
tapply(iris$Sepal.Length, iris$Species, mean)
```

```
#     setosa versicolor  virginica
#      5.006      5.936      6.588
```

```r
aggregate(Sepal.Length ~ Species, data=iris, mean)
```

```
#      Species Sepal.Length
# 1     setosa        5.006
# 2 versicolor        5.936
# 3  virginica        6.588
```

```r
aggregate(. ~ Species, data=iris, mean)
```

```
#      Species Sepal.Length Sepal.Width Petal.Length Petal.Width
# 1     setosa        5.006       3.428        1.462       0.246
# 2 versicolor        5.936       2.770        4.260       1.326
# 3  virginica        6.588       2.974        5.552       2.026
```

Ex 92: Memoization via Closure

Problem

- Write `make_memo(fn)` that returns a memoized version storing results in a list environment
- Memoize `fib`
- Show second call is instant

Sol 92

```r
make_memo <- function(fn) {
  cache <- list()
  function(...) {
    key <- paste(..., sep=",")
    if (is.null(cache[[key]])) cache[[key]] <<- fn(...)
    cache[[key]]
  }
}
slow_fn <- function(x) { Sys.sleep(0.1); x^2 }
fast_fn <- make_memo(slow_fn)
system.time(fast_fn(5))                          # slow
```

```
#    user  system elapsed
#     0.0     0.0     0.1
```

```r
system.time(fast_fn(5))                          # instant
```

```
#    user  system elapsed
#       0       0       0
```

Ex 93: `lapply` with Index

Problem

- Given `l <- list(a=1:3, b=4:6)`, use `lapply(seq_along(l), ...)` to create `"a: 1 2 3"`, `"b: 4 5 6"`
- Same with `Map` and `names(l)`
- Same with `mapply`

Sol 93

```r
l <- list(a=1:3, b=4:6)
lapply(seq_along(l), \(i)
  paste0(names(l)[i], ": ", paste(l[[i]], collapse=" ")))
```

```
# [[1]]
# [1] "a: 1 2 3"
#
# [[2]]
# [1] "b: 4 5 6"
```

```r
Map(\(nm, v) paste0(nm, ": ", paste(v,collapse=" ")),
    names(l), l)
```

```
# $a
# [1] "a: 1 2 3"
#
# $b
# [1] "b: 4 5 6"
```

```r
mapply(\(nm,v) paste0(nm,": ",paste(v,collapse=" ")),
       names(l), l)
```

```
#           a           b
```

Ex 94: Nested `lapply`

Problem

- Generate a 3×3 multiplication table using `lapply(1:3, \(i) sapply(1:3, \(j) i*j))`
- Convert to a matrix with `do.call(rbind, ...)`
- Compare with `outer(1:3, 1:3)`

Sol 94

```r
l <- lapply(1:3, \(i) sapply(1:3, \(j) i*j))
do.call(rbind, l)
```

```
#      [,1] [,2] [,3]
# [1,]    1    2    3
# [2,]    2    4    6
# [3,]    3    6    9
```

```r
outer(1:3, 1:3)                                    # same
```

```
#      [,1] [,2] [,3]
# [1,]    1    2    3
# [2,]    2    4    6
# [3,]    3    6    9
```

Ex 95: Map for Parallel Processing

Problem

`prices <- list(c(100,102,98), c(50,55,52), c(200,210,205)). names_v <- c("AAPL", "GOOG", "MSFT").`

- Compute return `diff(p)/p[-length(p)]` for each using `Map`
- Compute mean return per stock

Sol 95

```r
prices <- list(c(100,102,98), c(50,55,52), c(200,210,205))
names_v <- c("AAPL","GOOG","MSFT")
returns <- Map(\(p) diff(p)/p[-length(p)], prices)
names(returns) <- names_v
sapply(returns, mean)
```

```
#         AAPL         GOOG         MSFT
# -0.009607843  0.022727273  0.013095238
```

Ex 96: `Filter` + `Reduce` Combo

Problem

- From `1:100`, filter multiples of 3, then sum with `Reduce`
- From a list of vectors, keep only those with length > 3, then find their common intersection
- Compare with direct `sum(Filter(...))`

Sol 96

```r
Reduce("+", Filter(\(x) x%%3==0, 1:100))
```

```
# [1] 1683
```

```r
vecs <- list(1:10, 3:8, 2:15, 5:12, 4:20)
long <- Filter(\(v) length(v)>3, vecs)
Reduce(intersect, long)                          # 5:8
```

```
# [1] 5 6 7 8
```

```r
sum(Filter(\(x) x%%3==0, 1:100))                  # same as (a)
```

```
# [1] 1683
```

Strings and Regex (Ex 97–110)

Ex 97: Basic String Functions

Problem

`x <- c("  Hello World  ", "R Programming", "data SCIENCE").`

- Trim whitespace; count chars of each
- Convert to uppercase, lowercase
- `startsWith` / `endsWith` tests
- Extract first 4 chars using `substr`

Sol 97

```r
x <- c("  Hello World  ","R Programming","data SCIENCE")
xt <- trimws(x); nchar(xt)
```

```
# [1] 11 13 12
```

```r
toupper(xt); tolower(xt)
```

```
# [1] "HELLO WORLD"   "R PROGRAMMING" "DATA SCIENCE"
# [1] "hello world"   "r programming" "data science"
```

```r
startsWith(xt, "Hello"); endsWith(xt, "SCIENCE")
```

```
# [1]  TRUE FALSE FALSE
# [1] FALSE FALSE  TRUE
```

```r
substr(xt, 1, 4)
```

```
# [1] "Hell" "R Pr" "data"
```

Ex 98: paste and sprintf

Problem

- `paste("item", 1:5, sep="_")`
- Same with `collapse=", "`
- `sprintf("Name: %s, Score: %.1f", "Alice", 95.678)`
- Zero-pad: `sprintf("%03d", 1:5)`

Sol 98

```r
paste("item", 1:5, sep="_")
```

```
# [1] "item_1" "item_2" "item_3" "item_4" "item_5"
```

```r
paste("item", 1:5, sep="_", collapse=", ")
```

```
# [1] "item_1, item_2, item_3, item_4, item_5"
```

```r
sprintf("Name: %s, Score: %.1f", "Alice", 95.678)
```

```
# [1] "Name: Alice, Score: 95.7"
```

```r
sprintf("%03d", 1:5)
```

```
# [1] "001" "002" "003" "004" "005"
```

Ex 99: `strsplit`

Problem

- Split `"a.b.c.d"` by `"."` (note the escape!)
- Split `"hello world foo"` by space
- Split each character of `"ABCDE"`
- Split `c("a-b", "c-d-e")` — what shape is the result?

Sol 99

```r
strsplit("a.b.c.d", "\\.")
```

```
# [[1]]
# [1] "a" "b" "c" "d"
```

```r
strsplit("hello world foo", " ")
```

```
# [[1]]
# [1] "hello" "world" "foo"
```

```r
strsplit("ABCDE", "")
```

```
# [[1]]
# [1] "A" "B" "C" "D" "E"
```

```r
strsplit(c("a-b","c-d-e"), "-")                    # list of 2
```

```
# [[1]]
# [1] "a" "b"
#
# [[2]]
# [1] "c" "d" "e"
```

Ex 100: grep and grepl

Problem

```
w <- c("statistics","computation","algorithm","distribution","estimation").
```

- Words containing "tion"
- Words starting with vowel
- Words of ≥ 12 characters
- grepl to create logical mask

Sol 100

```r
w <- c("statistics","computation","algorithm",
       "distribution","estimation")
grep("tion", w, value=TRUE)
```

```
# [1] "computation"  "distribution" "estimation"
```

```r
grep("^[aeiou]", w, value=TRUE)
```

```
# [1] "algorithm"  "estimation"
```

```r
w[nchar(w) >= 12]
```

```
# [1] "distribution"
```

```r
grepl("tion", w)
```

```
# [1] FALSE  TRUE FALSE  TRUE  TRUE
```

Ex 101: `sub` and `gsub`

Problem

- Replace first space with `"_"` in `"hello world foo"`
- Replace all spaces
- Remove all digits from `"abc123def456"`
- Compress multiple spaces to one

Sol 101

```r
sub(" ", "_", "hello world foo")                    # first only
```

```
# [1] "hello_world foo"
```

```r
gsub(" ", "_", "hello world foo")                   # all
```

```
# [1] "hello_world_foo"
```

```r
gsub("[0-9]", "", "abc123def456")
```

```
# [1] "abcdef"
```

```r
gsub(" +", " ", "too   many    spaces")
```

```
# [1] "too many spaces"
```

Ex 102: Capture Groups

Problem

- Swap first/last name: `sub("(\\w+) (\\w+)", "\\2, \\1", "John Smith")`
- Extract domain from email: `sub(".*@", "", "user@example.com")`
- Extract year from dates using capture group

Sol 102

```r
sub("(\\w+) (\\w+)", "\\2, \\1",
    c("John Smith","Jane Doe"))
```

```
# [1] "Smith, John" "Doe, Jane"
```

```r
sub(".*@", "", c("user@example.com","a@b.org"))
```

```
# [1] "example.com" "b.org"
```

```r
sub("(\\d{4})-\\d{2}-\\d{2}", "\\1",
    c("2024-01-15","2023-12-31"))                    # year
```

```
# [1] "2024" "2023"
```

Ex 103: regexpr and gregexpr

Problem

`x <- "Phone: 02-1234-5678, Fax: 04-8765-4321".`

- Extract all phone numbers with `gregexpr` + `regmatches`
- Extract first match only with `regexpr`
- Extract all words with `\\b\\w+\\b`

Sol 103

```r
x <- "Phone: 02-1234-5678, Fax: 04-8765-4321"
m <- gregexpr("[0-9]+-[0-9]+-[0-9]+", x)
regmatches(x, m)
```

```
# [[1]]
# [1] "02-1234-5678" "04-8765-4321"
```

```r
m1 <- regexpr("[0-9]+-[0-9]+-[0-9]+", x)
regmatches(x, m1)
```

```
# [1] "02-1234-5678"
```

```r
regmatches(x, gregexpr("\\b\\w+\\b", x))
```

```
# [[1]]
# [1] "Phone" "02"    "1234"  "5678"  "Fax"   "04"    "8765"  "4321"
```

Ex 104: Greedy vs Lazy

Problem

`x <- "<b>bold</b> and <i>italic</i>".`

- `gregexpr("<.*>", x)` + `regmatches` — what matches? (greedy)
- `gregexpr("<.*?>", x)` — what matches? (lazy)
- Explain the difference

Sol 104

```r
x <- "<b>bold</b> and <i>italic</i>"
regmatches(x, gregexpr("<.*>", x))                  # one big match
```

```
# [[1]]
# [1] "<b>bold</b> and <i>italic</i>"
```

```r
regmatches(x, gregexpr("<.*?>", x))                 # 4 tags
```

```
# [[1]]
# [1] "<b>"  "</b>" "<i>"  "</i>"
```

```r
# greedy: .* matches as much as possible (all the way
#     to last >); lazy: .*? matches as little as possible
```

Ex 105: Anchors and Quantifiers

Problem

`x <- c("cat","concatenate","scat","catalog").`

- Starts with `cat`
- Ends with `cat`
- Exactly `cat`
- `c("a","ab","abb","abbb")`: match `ab+` vs `ab*` vs `ab?`

Sol 105

```r
x <- c("cat","concatenate","scat","catalog")
grep("^cat", x, value=TRUE)
```

```
# [1] "cat"     "catalog"
```

```r
grep("cat$", x, value=TRUE)
```

```
# [1] "cat"  "scat"
```

```r
grep("^cat$", x, value=TRUE)
```

```
# [1] "cat"
```

```r
y <- c("a","ab","abb","abbb")
grep("ab+", y, value=TRUE); grep("ab*", y, value=TRUE)
```

```
# [1] "ab"   "abb"  "abbb"
# [1] "a"    "ab"   "abb"  "abbb"
```

Ex 106: POSIX Classes

Problem

`x <- c("Hello!", "123", "abc", "A1!")`.

- Contains punctuation
- All digits
- Contains only alphabetic
- All alphanumeric

Sol 106

```r
x <- c("Hello!", "123", "abc", "A1!")
grep("[[:punct:]]", x, value=TRUE)
```

```
# [1] "Hello!" "A1!"
```

```r
grep("^[[:digit:]]+$", x, value=TRUE)
```

```
# [1] "123"
```

```r
grep("^[[:alpha:]]+$", x, value=TRUE)
```

```
# [1] "abc"
```

```r
grep("^[[:alnum:]]+$", x, value=TRUE)
```

```
# [1] "123" "abc"
```

Ex 107: Lookahead/Lookbehind

Problem

- Extract numbers preceded by $: `c("$100","200","$300")`
- Find files NOT ending in `.txt`: `c("a.txt","b.png","c.txt","d.jpg")`
- Explain why `perl=TRUE` is needed

Sol 107

```r
x <- c("$100","200","$300")
regmatches(x, gregexpr("(?<=\\$)\\d+", x, perl=TRUE))
```

```
# [[1]]
# [1] "100"
#
# [[2]]
# character(0)
#
# [[3]]
# [1] "300"
```

```r
y <- c("a.txt","b.png","c.txt","d.jpg")
grep("\\.(?!txt$)", y, value=TRUE, perl=TRUE)
```

```
# [1] "b.png" "d.jpg"
```

```r
# lookahead/lookbehind are Perl-compatible regex
# extensions not available in base (POSIX) regex
```

Ex 108: `regexec` — Structured Extraction

Problem

`dates <- c("2024-01-15 14:30", "2023-12-31 09:00")`.

- Write pattern to capture year, month, day, hour, minute separately
- Extract with `regexec` + `regmatches`
- Build a data frame from the results

Sol 108

```r
dates <- c("2024-01-15 14:30", "2023-12-31 09:00")
pat <- "(\\d{4})-(\\d{2})-(\\d{2}) (\\d{2}):(\\d{2})"
m <- regexec(pat, dates)
parts <- regmatches(dates, m)
do.call(rbind, lapply(parts, \(p)
  setNames(p[-1], c("Y","M","D","h","m"))))
```

```
#      Y      M    D    h    m
# [1,] "2024" "01" "15" "14" "30"
# [2,] "2023" "12" "31" "09" "00"
```

Ex 109: camelCase snake_case

Problem

- Write `camel_to_snake(s)` converting `"camelCase"` → `"camel_case"`
- Write `snake_to_camel(s)` converting `"snake_case"` → `"snakeCase"`
- Test both on `"myHTTPRequest"` / `"my_http_request"`

Sol 109

```r
camel_to_snake <- function(s) tolower(gsub("([A-Z])","_\\1",s))
camel_to_snake("camelCase")
```

```
# [1] "camel_case"
```

```r
camel_to_snake("myHTTPRequest")
```

```
# [1] "my_h_t_t_p_request"
```

```r
snake_to_camel <- function(s) {
  parts <- strsplit(s, "_")[[1]]
  paste0(parts[1], paste0(toupper(substring(parts[-1],1,1)),
         substring(parts[-1],2), collapse=""))
}
snake_to_camel("my_http_request")
```

```
# [1] "myHttpRequest"
```

Ex 110: Clean Column Names

Problem

`cols <- c("First Name", "Last.Name", "Age (years)", "Income$")`.

- Replace all non-alphanumeric chars with _
- Remove leading/trailing _
- Collapse consecutive _ to single _
- Convert to lowercase

Sol 110

```r
cols <- c("First Name","Last.Name","Age (years)","Income$")
x <- gsub("[^a-zA-Z0-9]", "_", cols)
x <- gsub("^_+|_+$", "", x)
x <- gsub("_+", "_", x)
tolower(x)
```

```
# [1] "first_name" "last_name"  "age_years"  "income"
```

Date/Time, I/O, Error Handling, Advanced (Ex 111–120)

Ex 111: Date Arithmetic

Problem

- Create today's date with `Sys.Date()`. Add 30 days
- Days between `2024-01-01` and `2024-12-31`
- Generate monthly sequence Jan–Jun 2024
- Extract weekday name and month name from a date

Sol 111

```r
today <- Sys.Date(); today + 30
```

```
# [1] "2026-04-22"
```

```r
as.Date("2024-12-31") - as.Date("2024-01-01")     # 365
```

```
# Time difference of 365 days
```

```r
seq(as.Date("2024-01-01"), as.Date("2024-06-01"),
    by="month")
```

```
# [1] "2024-01-01" "2024-02-01" "2024-03-01" "2024-04-01" "2024-05-01"
# [6] "2024-06-01"
```

```r
d <- as.Date("2024-07-15")
weekdays(d); months(d); format(d, "%A %B %d, %Y")
```

```
# [1] " "
# [1] " "
# [1] "    15, 2024"
```

Ex 112: `POSIXct` and Timezones

Problem

- Create `as.POSIXct("2024-01-15 14:30:00", tz="Asia/Taipei")`
- Extract hour, minute via `POSIXlt`
- Difference between two timestamps in hours
- Format with `strftime`

Sol 112

```r
t1 <- as.POSIXct("2024-01-15 14:30:00", tz="Asia/Taipei")
lt <- as.POSIXlt(t1); lt$hour; lt$min
```

```
# [1] 14
# [1] 30
```

```r
t2 <- as.POSIXct("2024-01-15 18:45:00", tz="Asia/Taipei")
difftime(t2, t1, units="hours")
```

```
# Time difference of 4.25 hours
```

```r
strftime(t1, "%Y/%m/%d %H:%M %Z")
```

```
# [1] "2024/01/15 14:30 CST"
```

Ex 113: Binary I/O

Problem

- `charToRaw("Hello")` — what type?
- Convert back with `rawToChar`
- `as.integer(charToRaw("ABC"))` — what are the byte values?
- Write raw bytes to a temp file with `writeBin` and read back

Sol 113

```r
r <- charToRaw("Hello"); r; typeof(r)              # raw
```

```
# [1] 48 65 6c 6c 6f
# [1] "raw"
```

```r
rawToChar(r)                                        # "Hello"
```

```
# [1] "Hello"
```

```r
as.integer(charToRaw("ABC"))                        # 65 66 67
```

```
# [1] 65 66 67
```

```r
tmp <- tempfile()
writeBin(charToRaw("test123"), tmp)
rawToChar(readBin(tmp, "raw", 20))
```

```
# [1] "test123"
```

Ex 114: `readline` and `interactive`

Problem

- Write a function that uses `readline` to ask for user's name (use `eval=FALSE`)
- Use `interactive()` to branch: interactive → readline, batch → default
- Why does `readline` always return character?

Sol 114

```r
ask_name <- function() {
  if (interactive()) {
    name <- readline("Your name: ")
  } else {
    name <- "default_user"
  }
  cat("Hello,", name, "\n")
}
# readline reads raw text from console; R has no way
# to know the intended type, so it always returns character
```

Ex 115: `system2`

Problem

- Run `system2("echo", args=c("hello","world"))`
- Capture output with `stdout=TRUE`
- Run in background with `wait=FALSE`
- Cross-platform check with `.Platform$OS.type`

Sol 115

```r
system2("echo", args=c("hello","world"))
out <- system2("date", stdout=TRUE); out
system2("sleep", args="1", wait=FALSE)
if (.Platform$OS.type == "unix") {
  system2("uname", args="-a", stdout=TRUE)
}
```

Ex 116: S3 Classes

Problem

- Create S3 class `Circle`: `list(radius=r)` with `class="Circle"`
- Write `print.Circle` and `area.Circle`
- Create instance; dispatch `print` and `area`

Sol 116

```r
Circle <- function(r) structure(list(radius=r), class="Circle")
print.Circle <- function(x, ...)
  cat(sprintf("Circle(r=%.2f)\n", x$radius))
area <- function(x, ...) UseMethod("area")
area.Circle <- function(x, ...) pi * x$radius^2
c1 <- Circle(5); c1
```

```
# Circle(r=5.00)
```

```r
area(c1)
```

```
# [1] 78.53982
```

Ex 117: Formula and `lm`

Problem

- `f <- y ~ x`; show `class()`, `typeof()`, `all.vars()`
- Fit `lm(dist ~ speed, data=cars)`. Extract coefficients
- Plot data with regression line

Sol 117

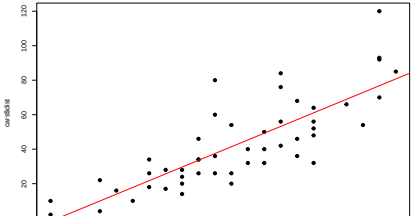
```r
f <- y ~ x; class(f); typeof(f); all.vars(f)
```

```
# [1] "formula"
# [1] "language"
# [1] "y" "x"
```

```r
model <- lm(dist ~ speed, data=cars)
coef(model)
```

```
# (Intercept)       speed
#  -17.579095    3.932409
```

```r
plot(cars$speed, cars$dist, pch=19)
abline(model, col="red", lwd=2)
```



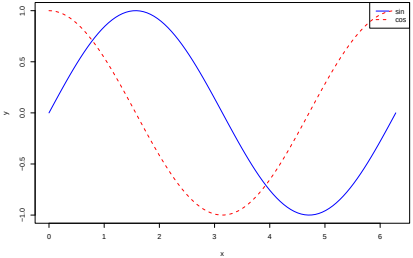
Ex 118: Plotting

Problem

- Plot `sin(x)` and `cos(x)` for `x` from 0 to 2π with legend
- `par(mfrow=c(1,2))`: histogram and boxplot side by side
- Add annotation with `text()` and horizontal line with `abline(h=...)`

Sol 118

```r
x <- seq(0, 2*pi, length=100)
plot(x, sin(x), type="l", col="blue", lwd=2, ylab="y")
lines(x, cos(x), col="red", lwd=2, lty=2)
legend("topright", c("sin","cos"), col=c("blue","red"),
       lty=1:2, lwd=2)
```



Ex 119: Random Number Generation

Problem

- `set.seed(42)` then generate 5 uniform, 5 normal, 10 Bernoulli
- Simulate 1000 dice rolls and tabulate
- Estimate π via Monte Carlo: sample `(x,y)` uniform in $[0,1]^2$, count proportion inside unit circle quadrant

Sol 119

```r
set.seed(42)
runif(5); rnorm(5); rbinom(10, 1, 0.5)
```

```
# [1] 0.9148060 0.9370754 0.2861395 0.8304476 0.6417455
# [1]  0.04788474 -1.10459944  0.53902380  0.58020632 -0.65750284
#  [1] 1 1 0 0 1 1 0 1 1 0
```

```r
table(sample(1:6, 1000, replace=TRUE))
```

```
#
#   1   2   3   4   5   6
# 169 191 163 157 153 167
```

```r
n <- 1e5; x <- runif(n); y <- runif(n)
4 * mean(x^2 + y^2 <= 1)  #
```

```
# [1] 3.14176
```

Ex 120: Options and Benchmarking

Problem

- Save current `digits` option, set to 15, print `pi`, restore
- Time `crossprod(x)` for `x <- matrix(rnorm(1e6),1000)` with `system.time`
- Use `proc.time` for manual timing of `sum(1:1e7)`

Sol 120

```r
old <- options(digits=15); pi
```

```
# [1] 3.14159265358979
```

```r
options(old)
system.time({
  x <- matrix(rnorm(1e6), 1000, 1000)
  y <- crossprod(x)
})
```

```
#    user  system elapsed
#   0.099   0.070   0.038
```

```r
pt <- proc.time(); s <- sum(1:1e7); proc.time()-pt
```

```
#    user  system elapsed
#   0.012   0.002   0.001
```

Closures and Lexical Scoping (Ex 121–128)

Ex 121: Power Factory

Problem

- Write `power_factory(n)` that returns a function computing x^n
- Create `square` and `cube` from it; test on 5
- Inspect the captured `n` with `environment(square)$n`
- Verify `square` and `cube` have different environments

Sol 121

```r
power_factory <- function(n) function(x) x^n
square <- power_factory(2); cube <- power_factory(3)
square(5); cube(5)
```

```
# [1] 25
# [1] 125
```

```r
environment(square)$n; environment(cube)$n          # 2; 3
```

```
# [1] 2
# [1] 3
```

```r
identical(environment(square), environment(cube))   # FALSE
```

```
# [1] FALSE
```

Ex 122: Closure Counter

Problem

Write `make_counter()` returning `list(increment, get, reset)`.

- `increment()` adds 1 to internal `n` and returns it
- `get()` returns current `n`; `reset()` sets `n` back to 0
- Create two independent counters; show they don't share state
- Rewrite using `local()` for a single-instance counter

Sol 122

```r
make_counter <- function() {
  n <- 0
  list(increment = function() { n <<- n + 1; n },
       get = function() n,
       reset = function() { n <<- 0 })
}
c1 <- make_counter(); c2 <- make_counter()
c1$increment(); c1$increment(); c2$increment()
```

```
# [1] 1
# [1] 2
# [1] 1
```

```r
cat("c1:", c1$get(), " c2:", c2$get(), "\n")
```

```
# c1: 2  c2: 1
```

```r
c1$reset(); c1$get()
```

```
# [1] 0
```

```r
# local() single-instance
counter <- local({
```

Ex 123: Loop Capture Bug

Problem

- Create 5 functions in a `for` loop: `funs[[i]] <- function() i`. Call all 5. What happens?
- Fix using `force()` inside a wrapper function
- Fix using `lapply`
- Explain *why* the bug occurs (lazy evaluation + shared environment)

Sol 123

```r
funs <- list()
for (i in 1:5) funs[[i]] <- function() i
sapply(funs, \(f) f())                                  # all 5!
```

```
# [1] 5 5 5 5 5
```

```r
# Bug: all closures share the same env; i=5 after loop

make_fun <- function(val) { force(val); function() val }
funs <- list()
for (i in 1:5) funs[[i]] <- make_fun(i)
sapply(funs, \(f) f())
```

```
# [1] 1 2 3 4 5
```

```r
funs <- lapply(1:5, \(i) function() i)
sapply(funs, \(f) f())
```

```
# [1] 1 2 3 4 5
```

```r
# Lazy eval: function() i captures a promise to i,
# not its value. By call time, loop finished, i=5.
```

Ex 124: Memoized Fibonacci

Problem

Write `make_memo_fib()` returning `list(fib, cache_size, clear_cache)`.

- Use a named list as cache; `fib(n)` checks cache before computing
- Time `fib(35)` first call vs second call
- How large is the cache after `fib(35)`?
- Clear cache and verify `fib(10)` still works

Sol 124

```r
make_memo_fib <- function() {
  cache <- list()
  fib <- function(n) {
    key <- as.character(n)
    if (!is.null(cache[[key]])) return(cache[[key]])
    result <- if (n <= 2) 1 else fib(n-1) + fib(n-2)
    cache[[key]] <<- result; result
  }
  list(fib = fib,
       cache_size = function() length(cache),
       clear_cache = function() { cache <<- list() })
}
mf <- make_memo_fib()
system.time(mf$fib(35))                              # first
```

```
#    user  system elapsed
#   0.000   0.000   0.001
```

```r
system.time(mf$fib(35))                              # second: instant
```

```
#    user  system elapsed
#       0       0       0
```

Ex 125: Decorator — Logging Wrapper

Problem

Write `with_logging(f)` that wraps `f` to print call count on each invocation.

- Return a closure that increments a counter and calls `f(...)`
- Test: `logged_mean <- with_logging(mean)`. Call it 3 times
- Extend to also print elapsed time per call

Sol 125

```r
with_logging <- function(f, fname = deparse(substitute(f))) {
  count <- 0L
  function(...) {
    count <<- count + 1L
    t0 <- proc.time()["elapsed"]
    cat(sprintf("[%s] call #%d\n", fname, count))
    result <- f(...)
    cat(sprintf("  elapsed: %.4fs\n",
        proc.time()["elapsed"] - t0))
    result
  }
}
logged_mean <- with_logging(mean)
logged_mean(1:100); logged_mean(rnorm(1e5))
```

```
# [mean] call #1
#   elapsed: 0.0000s
```

```
# [1] 50.5
```

```
# [mean] call #2
#   elapsed: 0.0030s
```

Ex 126: Bank Account with Transaction Log

Problem

Write `make_account(owner, balance)` returning `list(deposit, withdraw, statement, balance)`.

- `deposit(amount)` adds and logs. `withdraw(amount)` subtracts; `stop()` if insufficient
- `statement()` returns a data frame of all transactions
- Create two accounts; verify independence

Sol 126

```r
make_account <- function(owner, init = 0) {
  bal <- init
  log <- data.frame(time=Sys.time()[0], type=character(0),
                    amount=numeric(0), balance=numeric(0))
  record <- function(type, amount) {
    log <<- rbind(log, data.frame(
      time=Sys.time(), type=type, amount=amount, balance=bal))
  }
  list(
    deposit = function(amt) {
      stopifnot(amt > 0); bal <<- bal + amt
      record("deposit", amt); invisible(bal)
    },
    withdraw = function(amt) {
      stopifnot(amt > 0)
      if (amt > bal) stop("Insufficient funds")
      bal <<- bal - amt; record("withdraw", amt); invisible(bal)
    },
    balance = function() bal,
    statement = function() log)
}
```

Ex 127: Moving Average Filter

Problem

Write `make_ema(alpha)` returning a function that computes an exponential moving average: $\text{EMA}_t = \alpha x_t + (1-\alpha)\text{EMA}_{t-1}$.

- First call initializes EMA to the input value
- Test with prices `c(100, 102, 101, 105, 103, 107, 106)` and $\alpha = 0.2$
- Create two EMAs with different α; verify independence

Sol 127

```r
make_ema <- function(alpha = 0.3) {
  value <- NULL
  function(x) {
    if (is.null(value)) value <<- x
    else value <<- alpha * x + (1 - alpha) * value
    value
  }
}
ema <- make_ema(0.2)
sapply(c(100, 102, 101, 105, 103, 107, 106), ema)
```

```
# [1] 100.0000 100.4000 100.5200 101.4160 101.7328 102.7862 103.4290
```

```r
ema_fast <- make_ema(0.5); ema_slow <- make_ema(0.1)
sapply(c(100, 110, 90), ema_fast)
```

```
# [1] 100.0 105.0  97.5
```

```r
sapply(c(100, 110, 90), ema_slow)
```

```
# [1] 100.0 101.0  99.9
```

Ex 128: Fibonacci Iterator

Problem

Write `make_fib_iterator()` returning `list(next_val, reset)`.

- `next_val()` returns the next Fibonacci number on each call
- Use `replicate(12, fib_it$next_val())` to get first 12 values
- Reset and verify the sequence restarts

Sol 128

```r
make_fib_iterator <- function() {
  a <- 0; b <- 1
  list(
    next_val = function() {
      val <- a; new <- a + b; a <<- b; b <<- new; val
    },
    reset = function() { a <<- 0; b <<- 1 })
}
fib_it <- make_fib_iterator()
replicate(12, fib_it$next_val())
```

```
#  [1]  0  1  1  2  3  5  8 13 21 34 55 89
```

```r
fib_it$reset(); replicate(5, fib_it$next_val())
```

```
# [1] 0 1 1 2 3
```

Split–Apply–Combine Deep Dive (Ex 129–136)

Ex 129: Multi-Factor Split and Unsplit

Problem

Using `ToothGrowth`:

- Split by both `supp` and `dose`; how many groups?
- Compute mean `len` per group using `sapply`
- Add a z-score column within each group using `split` + `lapply` + `do.call(rbind,...)`
- Use `unsplit` to restore original ordering after modifying split pieces

Sol 129

```r
tg <- ToothGrowth
sp <- split(tg, list(tg$supp, tg$dose))
length(sp)                                          # 6
```

```
# [1] 6
```

```r
sapply(sp, \(d) mean(d$len))
```

```
# OJ.0.5 VC.0.5   OJ.1   VC.1   OJ.2   VC.2
#  13.23   7.98  22.70  16.77  26.06  26.14
```

```r
parts <- lapply(sp, \(d) {
  d$len_z <- (d$len - mean(d$len)) / sd(d$len); d })
tg2 <- do.call(rbind, parts)
head(tg2[order(as.numeric(rownames(tg2))), ], 4)
```

```
#            len supp dose      len_z
# OJ.0.5.31 15.2   OJ  0.5  0.4417329
# OJ.0.5.32 21.5   OJ  0.5  1.8543813
# OJ.0.5.33 17.6   OJ  0.5  0.9798847
# OJ.0.5.34  9.7   OJ  0.5 -0.7915316
```

```r
# unsplit example
```

Ex 130: `aggregate()` — Matrix Column Gotcha

Problem

Using `ToothGrowth`:

- Use `aggregate` with `FUN = function(x) c(m=mean(x), s=sd(x))`. What type is the result column?
- Expand the matrix column into a proper flat data frame
- Use `cbind(mpg, hp, wt) ~ cyl` to summarize multiple columns of `mtcars`
- Use `.` shorthand: `aggregate(. ~ cyl, ...)` on a subset of `mtcars`

Sol 130

```r
tg <- ToothGrowth
res <- aggregate(len ~ supp, data = tg,
  FUN = function(x) c(m=mean(x), s=sd(x)))
str(res)                                                # matrix col!
```

```
# 'data.frame': 2 obs. of  2 variables:
#  $ supp: Factor w/ 2 levels "OJ","VC": 1 2
#  $ len : num [1:2, 1:2] 20.66 16.96 6.61 8.27
#   ..- attr(*, "dimnames")=List of 2
#   .. ..$ : NULL
#   .. ..$ : chr [1:2] "m" "s"
```

```r
cbind(res[1], as.data.frame(res$len))                   # flatten
```

```
#   supp        m        s
# 1   OJ 20.66333 6.605561
# 2   VC 16.96333 8.266029
```

```r
aggregate(cbind(mpg, hp, wt) ~ cyl,
          data = mtcars, FUN = mean)
```

```
#   cyl      mpg        hp       wt
# 1   4 26.66364  82.63636 2.285727
```

Ex 131: `merge()` — All Join Types

Problem

`cust <- data.frame(id=1:4, name=c("A","B","C","D")). ord <- data.frame(cid=c(2,3,3,5), prod=c("pen","book","ink","tape")).`

- Inner join. Which rows are dropped and why?
- Left join. What appears for customer D?
- Right join. What appears for `cid=5`?
- Full outer join

Sol 131

```r
cust <- data.frame(id=1:4, name=c("A","B","C","D"))
ord <- data.frame(cid=c(2,3,3,5),
                  prod=c("pen","book","ink","tape"))
merge(cust, ord, by.x="id", by.y="cid")              # inner
```

```
#   id name prod
# 1  2    B  pen
# 2  3    C book
# 3  3    C  ink
```

```r
merge(cust, ord, by.x="id", by.y="cid", all.x=TRUE)  # left
```

```
#   id name prod
# 1  1    A <NA>
# 2  2    B  pen
# 3  3    C book
# 4  3    C  ink
# 5  4    D <NA>
```

```r
merge(cust, ord, by.x="id", by.y="cid", all.y=TRUE)  # right
```

```
#   id name prod
# 1  2    B  pen
```

Ex 132: Multi-Table Merge with `Reduce`

Problem

- Create `products(pid, name)`, `orders(oid, pid, qty)`, `prices(pid, unit_price)`
- Inner-join all three; compute `total = qty * unit_price`
- Add `regions(pid, region)`. Use `Reduce` to merge all four

Sol 132

```r
products <- data.frame(pid=1:5,
  name=c('Pen','Book','Ink','Paper','Tape'))
orders <- data.frame(oid=101:106,
  pid=c(2,1,3,2,5,1), qty=c(3,10,2,5,1,8))
prices <- data.frame(pid=1:5,
  unit_price=c(1.5, 12, 8, 3, 2.5))
m <- merge(merge(products, orders, by='pid'),
           prices, by='pid')
m$total <- m$qty * m$unit_price; m[, c('oid','name','total')]
```

```
#   oid name total
# 1 102  Pen  15.0
# 2 106  Pen  12.0
# 3 101 Book  36.0
# 4 104 Book  60.0
# 5 103  Ink  16.0
# 6 105 Tape   2.5
```

```r
regions <- data.frame(pid=1:5, region=c('N','S','N','S','N'))
Reduce(\(x, y) merge(x, y, by='pid', all=TRUE),
       list(products, orders, prices, regions))
```

Ex 133: vapply vs sapply Safety

Problem

- `sapply(list(a=1:3, b=4:6), identity)` returns a matrix. `sapply(list(a=1:3, b=4:5), identity)` returns a list. Demonstrate
- Show that `vapply(list(a=1:3, b=4:6), identity, numeric(1))` throws an error
- Use `vapply` correctly with `FUN.VALUE = numeric(3)` for same-length vectors
- Use `vapply(iris[,1:4], median, numeric(1))` to safely compute medians

Sol 133

```r
sapply(list(a=1:3, b=4:6), identity)                # matrix
```

```
#      a b
# [1,] 1 4
# [2,] 2 5
# [3,] 3 6
```

```r
sapply(list(a=1:3, b=4:5), identity)                # list!
```

```
# $a
# [1] 1 2 3
#
# $b
# [1] 4 5
```

```r
vapply(list(a=1:3, b=4:6), identity, numeric(1))    # error!
```

```
# Error in vapply(list(a = 1:3, b = 4:6), identity, numeric(1)): values must be length 1,
#  but FUN(X[[1]]) result is length 3
```

```r
vapply(list(a=1:3, b=4:6), identity, numeric(3))    # OK
```

```
#      a b
# [1,] 1 4
```

Ex 134: `rapply()` — Recursive Apply

Problem

`nested <- list(a=1:3, b=list(c=4:6, d=list(e=7:9, f="text")))`.

- Use `rapply` with `sum`, `classes="integer"` to sum only integer leaves
- Same with `how="list"` to preserve structure
- Use `how="replace"` to multiply all integer leaves by 10
- What happens to the `"text"` element in each case?

Sol 134

```r
nested <- list(a=1:3,
  b=list(c=4:6, d=list(e=7:9, f="text")))
rapply(nested, sum, classes="integer")                # unlist
```

```
#     a   b.c b.d.e
#     6    15    24
```

```r
rapply(nested, sum, classes="integer", how="list")   # structure
```

```
# $a
# [1] 6
#
# $b
# $b$c
# [1] 15
#
# $b$d
# $b$d$e
# [1] 24
#
# $b$d$f
# NULL
```

Ex 135: `Reduce` — Running Max Drawdown

Problem

`r <- c(0.02, -0.01, 0.03, -0.04, 0.01, -0.02, 0.05)`.

- Use `Reduce` with `accumulate=TRUE` to compute cumulative wealth $W_t = W_{t-1}(1 + r_t)$, starting from $W_0 = 1$
- Compute running max of cumulative wealth using `Reduce` and `max`
- Compute drawdown: $\text{DD}_t = 1 - W_t / \max_{s \le t} W_s$
- Verify with vectorized `1 - cumprod(1+r) / cummax(cumprod(1+r))`

Sol 135

```r
r <- c(0.02, -0.01, 0.03, -0.04, 0.01, -0.02, 0.05)
W <- Reduce(\(w, ri) w*(1+ri), r, init=1,
            accumulate=TRUE)[-1]
Wmax <- Reduce(\(mx, w) max(mx, w), W,
               accumulate=TRUE)
DD <- 1 - W / Wmax; round(DD, 6)
```

```
# [1] 0.000000 0.010000 0.000000 0.040000 0.030400 0.049792 0.002282
```

```r
DD2 <- 1 - cumprod(1+r) / cummax(cumprod(1+r))
all.equal(DD, DD2)                                  # TRUE
```

```
# [1] TRUE
```

Ex 136: `Filter` Pipeline with |>

Problem

`words <- c("apple","Banana","cherry","Date","123","fig","GRAPE","hi")`.

- Use `Filter` to keep only all-lowercase words
- Use `Filter` + `Negate` to remove strings containing digits
- Find first word with > 5 characters using `Find`; find its position with `Position`
- Chain with |>: filter lowercase $\rightarrow$ filter length > 3 $\rightarrow$ sort

Sol 136

```r
words <- c("apple","Banana","cherry","Date","123",
           "fig","GRAPE","hi")
Filter(\(w) w == tolower(w), words)
```

```
# [1] "apple"  "cherry" "123"    "fig"    "hi"
```

```r
Filter(Negate(\(w) grepl("[0-9]", w)), words)
```

```
# [1] "apple"  "Banana" "cherry" "Date"   "fig"    "GRAPE"  "hi"
```

```r
Find(\(w) nchar(w) > 5, words)
```

```
# [1] "Banana"
```

```r
Position(\(w) nchar(w) > 5, words)
```

```
# [1] 2
```

```r
words |>
  Filter(f = \(w) w == tolower(w)) |>
  Filter(f = \(w) nchar(w) > 3) |>
  sort()
```

```
# [1] "apple"  "cherry"
```

Advanced Closure Applications (Ex 137–140)

Ex 137: Closure-Based ODE Solver (Euler Method)

Problem

Write `make_ode_solver(f, y0, dt)` where $f(t, y)$ defines dy/dt.

- `step()` advances one time step; `run(n)` returns trajectory as a data frame
- Solve $dy/dt = -0.5y$, $y(0) = 10$, $dt = 0.1$ for 100 steps
- Plot Euler vs exact $y = 10e^{-0.5t}$

Sol 137

```r
make_ode_solver <- function(f, y0, dt = 0.01) {
  t <- 0; y <- y0
  list(
    run = function(n) {
      traj <- matrix(NA, n+1, 2, dimnames=list(NULL, c("t","y")))
      traj[1, ] <- c(t, y)
      for (i in seq_len(n)) {
        y <<- y + f(t, y)*dt; t <<- t + dt
        traj[i+1, ] <- c(t, y)
      }
      as.data.frame(traj)
    },
    state = function() c(t=t, y=y),
    reset = function(y0_new=y0) { t <<- 0; y <<- y0_new })
}
solver <- make_ode_solver(\(t, y) -0.5*y, y0=10, dt=0.1)
traj <- solver$run(100)
plot(traj$t, traj$y, type='l', lwd=2, col='steelblue',
     xlab='t', ylab='y', main='Euler vs Exact')
curve(10*exp(-0.5*x), add=TRUE, col='red', lwd=2, lty=2)
```

Ex 138: Rate Limiter

Problem

Write `make_rate_limiter(f, max_calls, period)` wrapping `f` so at most `max_calls` invocations are allowed per `period`-second window.

- If limit exceeded, return `NULL` with a warning
- Track timestamps in a closure variable
- Test: limiter allowing 3 calls per 2 seconds; call 5 times rapidly

Sol 138

```r
make_rate_limiter <- function(f, max_calls=3, period=2) {
  timestamps <- numeric(0)
  function(...) {
    now <- as.numeric(Sys.time())
    timestamps <<- timestamps[timestamps > now - period]
    if (length(timestamps) >= max_calls) {
      warning("Rate limit exceeded"); return(NULL)
    }
    timestamps <<- c(timestamps, now)
    f(...)
  }
}
limited_sqrt <- make_rate_limiter(sqrt, 3, 2)
for (i in 1:5) {
  r <- limited_sqrt(i * 10)
  cat(sprintf("Call %d: %s\n", i,
      if (is.null(r)) "BLOCKED" else sprintf("%.4f", r)))
}
```

```
# Call 1: 3.1623
# Call 2: 4.4721
```

Ex 139: Portfolio Analytics Factory

Problem

Write `make_portfolio(weights, tickers)` returning:

- `returns(price_matrix)` — weighted portfolio returns
- `sharpe(price_matrix, rf)` — annualized Sharpe ratio
- `drawdown(price_matrix)` — maximum drawdown
- Test with simulated 252-day prices for 3 assets

Sol 139

```r
make_portfolio <- function(w, tickers) {
  w <- w / sum(w); names(w) <- tickers
  list(
    returns = function(P) {
      r <- apply(P, 2, \(x) diff(x)/head(x,-1)); drop(r %*% w)
    },
    sharpe = function(P, rf=0) {
      r <- apply(P, 2, \(x) diff(x)/head(x,-1))
      pr <- drop(r %*% w)
      (mean(pr) - rf/252) / sd(pr) * sqrt(252)
    },
    drawdown = function(P) {
      r <- apply(P, 2, \(x) diff(x)/head(x,-1))
      cum <- cumprod(1 + drop(r %*% w))
      max(1 - cum / cummax(cum))
    })
}
set.seed(42); P <- matrix(NA, 253, 3); P[1,] <- c(100,50,200)
for (i in 2:253) P[i,] <- P[i-1,]*exp(rnorm(3, 3e-4,
                                            c(.015,.02,.01)))
```

Ex 140: `by()` and Full EDA Report

Problem

Using `mtcars`:

- Use `by(mtcars, mtcars$cyl, \(d) cor(d$mpg, d$wt))` to compute per-cylinder correlation
- Mean, sd, min, max of `mpg` by `cyl` in one `aggregate` call; expand the matrix column
- Merge with group counts from `aggregate(..., FUN=length)`
- Sort by mean `mpg` descending

Sol 140

```r
by(mtcars, mtcars$cyl, \(d) cor(d$mpg, d$wt))
```

```
# mtcars$cyl: 4
# [1] -0.7131848
# ------------------------------------------------------------
# mtcars$cyl: 6
# [1] -0.6815498
# ------------------------------------------------------------
# mtcars$cyl: 8
# [1] -0.650358
```

```r
res <- aggregate(mpg ~ cyl, data=mtcars,
  FUN=\(x) c(mean=mean(x), sd=sd(x), min=min(x), max=max(x)))
flat <- cbind(res[1], as.data.frame(res$mpg))

counts <- aggregate(mpg ~ cyl, data=mtcars, FUN=length)
names(counts)[2] <- 'n'
flat <- merge(flat, counts, by='cyl')
flat[order(-flat$mean), ]
```

```
#   cyl     mean       sd  min  max  n
# 1   4 26.66364 4.509828 21.4 33.9 11
```